ASTRA: Autonomous Scientific Discovery in Astrophysics

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ABSTRACT

We present Astra (Autonomous System for Scientific Discovery in Astrophysics), an integrated framework that unifies numerical data analysis, causal reasoning, and physical validation within a single reproducible pipeline. Astra chains dimensional analysis via the Buckingham π theorem, structural causal models, Bayesian evidence computation, and multi-wavelength data fusion into coherent, physics-aware workflows. A cognitive architecture incorporating a dynamic knowledge graph, neuro-symbolic reasoning, and meta-cognitive monitoring coordinates the analytical loop, while a multi-agent scientific debate system evaluates competing hypotheses through structured argumentation and consensus mechanisms. A stigmergic swarm intelligence layer—inspired by ant colony foraging—coordinates hypothesis exploration through five specialized agents that deposit and follow digital pheromone trails. We validate the framework through six controlled astrophysical test cases, six live deployments recovering established physics (including Kepler's third law at $R^2 = 0.9982$), and cross-domain validation across economics, climate science, and epidemiology, collectively validating 38 hypotheses across five scientific domains. Astra is a tool to assist scientists, not replace domain expertise. Code and documentation: https://github.com/Tilanthi/ASTRA.

Key words: methods: data analysis – methods: statistical – methods: general – techniques: miscellaneous – ISM: clouds – stars: formation

1 INTRODUCTION

Modern astronomical surveys generate petabytes of multi-wavelength data per year, and a rich ecosystem of computational tools—from classifiers to nested-sampling codes, from causal discovery algorithms (Spirtes et al. 2000) to frontier large language models—gives astronomers unprecedented analytical power. Yet these tools operate largely in isolation: astronomers must manually stitch together separate pipelines for data processing, statistical modelling, causal inference, and physical validation, creating opportunities for inconsistent treatment of uncertainties, missed physical constraints, and analyses that are difficult to reproduce.

Astra (Autonomous System for Scientific Discovery in Astrophysics) addresses this integration challenge. Rather than replacing any single tool, Astra provides a unified framework that chains established algorithms into coherent, physics-aware analytical workflows. The system combines:

- Numerical data processing and statistical analysis
- Causal reasoning with structural causal models
- Physical validation through dimensional analysis and conservation laws
- Multi-wavelength data fusion with uncertainty propagation
- Hypothesis generation from pattern recognition
- Bayesian model selection with evidence computation
- Stigmergic swarm intelligence for autonomous exploration–exploitation balancing
- Cognitive architecture with dynamic knowledge graph and neuro-symbolic reasoning
- Multi-agent scientific debate for hypothesis evaluation and consensus
- Theory framework with automated consistency checking and abductive reasoning

This integrated approach ensures that physical validation, causal reasoning, and uncertainty quantification are applied consistently throughout multi-step analyses. While individual components use established algorithms (PC algorithm for causal discovery, nested sampling for Bayesian evidence, the Buckingham π theorem for dimensional analysis; Buckingham 1914), their integration within a single framework enables analyses that would otherwise require manual combination of multiple separate tools.

In this paper, we focus on six examples demonstrating how this integrated approach can be applied to astrophysical inference problems:

(i) **Scaling Relations Analysis** (Section 3): Dimensional analysis and physical validation of filament scaling relations

(ii) **Multi-Wavelength Data Fusion** (Section 4): Probabilistic cross-matching with Bayesian uncertainty propagation

(iii) **Pattern Recognition** (Section 5): Identification of established galaxy property relationships

(iv) **Causal Inference** (Section 6): Structural causal model discovery in stellar data

(v) **Bayesian Model Selection** (Section 7): Evidence-based model comparison with prior specification

(vi) **Discovery-Mode Operation on Synthetic Data** (Section 8):

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Knowledge-isolated pattern discovery and causal inference on the star formation threshold problem

The first five test cases use real observational data to validate Astra's ability to correctly recover known astrophysical results. The sixth test case demonstrates discovery-mode operation beyond knowledge retrieval, under controlled conditions. Using synthetic data with embedded causal structure (ground truth known to the author but not to Astra), it operates in "knowledge isolation mode" to discover patterns without being told what to look for, and generates testable predictions for future validation.

Section 9 presents six live deployments on archival data that independently recover established physics. To test whether Astra's analytical capabilities generalize beyond astrophysics, Section 10 presents cross-domain validation across economics, climate science, and epidemiology using public data from the World Bank, NASA GISS, and NOAA. Section 11 discusses why these results demonstrate Astra's unique capabilities and clarifies Astra's role as a tool to assist rather than replace scientists, and Section 12 concludes.

2 METHODS: ASTRA SYSTEM ARCHITECTURE

2.1 Overview

Astra implements a modular architecture designed for astrophysical data analysis and inference. The system integrates multiple specialized components through a coordinated framework, enabling complex multi-step analyses that combine data processing, physical reasoning, causal inference, and statistical validation.

2.1.1 Scope and Limitations

What Astra Demonstrates: The six astrophysical test cases show that Astra can: discover physical laws from data with theoretical validation; fuse multi-wavelength observations with proper uncertainty handling; identify established astrophysical relationships through pattern recognition; distinguish causation from correlation; perform rigorous Bayesian model comparison; and operate in knowledge isolation mode to discover patterns and generate testable predictions beyond knowledge retrieval. The cross-domain validation (Section 10) further demonstrates that these capabilities generalize to economics, climate science, and epidemiology, recovering approximately 42 validated hypotheses across five scientific domains.

What Astra Does Not Claim: Astra is not presented as achieving artificial general intelligence or AGI-like performance. The system operates within defined astrophysical domains using established algorithms (PC algorithm, Bayesian inference, dimensional analysis via the Buckingham π theorem, FCI causal discovery; Zhang 2008) combined through an integrated architecture. Results are validated against known physical theory and observational constraints. The system does not claim general reasoning beyond its training domains or autonomous operation without human oversight; the live dashboard (Section 2.6) provides real-time safety monitoring and intervention controls. The architecture supports natural language queries that trigger autonomous task completion, though this capability is not demonstrated in the present test cases. A stigmergic swarm intelligence layer (Section 2.7) coordinates autonomous exploration, while the live dashboard (Section 2.6) provides real-time monitoring and intervention controls. The cognitive architecture (Section 2.8) and multi-agent debate system (Section 2.9) are early-stage components whose integration is described but not independently validated in the present test cases.

Astra's Role: Astra is a tool to assist the astronomer, not a replacement for domain expertise. Its discovery architecture, used alongside an experienced scientist, provides capabilities in discovery and inference that go beyond those of straightforward AI or machine learning alone, as discussed in Section 11.6.

Validation Scope: The first five test cases use real observational data to validate Astra's ability to correctly recover known astrophysical results. The sixth test case demonstrates discovery capability on synthetic data with known ground truth, with testable predictions that require future observational validation. The cross-domain validation (Section 10) extends the scope to five scientific domains using exclusively public data from established archives. Generalization to additional domains or datasets requires further validation.

2.2 Core Architectural Components

(i) **Data Processing Pipeline:** Astra accepts heterogeneous astronomical data formats including catalogues (CSV, FITS), time-series data, images, and spectral data. The pipeline performs ingestion, validation, cleaning, and normalization while preserving measurement uncertainties and metadata.

(ii) **Physics Engine:** A unified physics engine implements fundamental physical laws and constraints including conservation laws (mass, energy, momentum), dimensional analysis using the Buckingham π theorem (Buckingham 1914), equation solving and numerical simulation, and units consistency checking throughout calculations. This enables Astra to validate discoveries against first principles rather than treating correlations as sufficient explanations.

(iii) **Causal Reasoning Module:** The causal reasoning module discovers and analyses causal structures using the PC algorithm (Spirtes et al. 2000) for learning directed acyclic graphs from observational data, the FCI algorithm (Zhang 2008) for causal discovery in the presence of latent confounders, conditional independence testing for continuous variables (Fisher's Z-test), V-structure detection for identifying colliders, do-calculus (Pearl 2009) for predicting effects of interventions, and domain-specific adaptations for astrophysical contexts. This module enables Astra to distinguish causal relationships from correlations, addressing a fundamental limitation of both traditional ML and LLM approaches.

(iv) **Bayesian Inference Engine:** Rigorous model comparison and uncertainty quantification are performed through evidence computation via marginal likelihood estimation, learned harmonic mean estimation (Spurio Mancini et al. 2023), Bayes factors and PSIS-LOO-CV for model comparison (Eadie et al. 2023), automatic complexity penalty (Occam's razor), posterior predictive checking, and Monte Carlo methods for uncertainty propagation. Model complexity is assessed using the Bayesian Information Criterion (BIC; Schwarz 1978) as a rapid approximation where full evidence computation is unnecessary.

(v) **Domain Knowledge Systems:** Organized astrophysical knowledge through MORK ontology encoding taxonomic and causal relationships, knowledge graphs for semantic relationships, vector-based semantic memory for concept retrieval, and 75 specialized domain modules (ISM, star formation, exoplanets, etc.).

(vi) **Multi-Module Orchestration:** Coordinates different analytical perspectives through specialized processing modules for different analysis types, a meta-cognitive layer for context-appropriate module selection, result synthesis and conflict resolution, and quality control and validation checks.

2.3 Unique Capabilities

Astra integrates sixteen distinct analytical capabilities organized into four groups:

Causal and Statistical Analysis: (1) *Bias Detection*—identifies and quantifies observational biases from survey geometry and selection effects. (2) *Scaling Relations Discovery*—discovers physical laws through dimensional analysis and functional form detection. (3) *Causal Inference*—discovers causal structures using the PC and FCI algorithms (Spirtes et al. 2000; Zhang 2008) and distinguishes physical laws from selection biases. (4) *Model Selection*—compares competing theories using Bayesian evidence with proper complexity penalties.

Data Integration and Analysis: (5) *Multi-Wavelength Fusion*—cross-matches sources across wavelengths with astrometric uncertainty propagation. (6) *Uncertainty Quantification*—propagates measurement errors through first-order and Monte Carlo methods. (7) *Temporal Reasoning*—detects periodic signals and forecasts time-series behaviour. (8) *Instrument-Aware Analysis*—evaluates observational requirements across astronomical facilities.

Knowledge Generation: (9) *Hypothesis Generation*—identifies patterns, detects anomalies, and generates testable hypotheses. (10) *Analogical Reasoning*—discovers structural mappings between different astrophysical systems. (11) *Counterfactual Analysis*—simulates physically-grounded scenarios to predict intervention effects. (12) *Physical Model Discovery*—identifies functional forms and validates against theoretical predictions.

Meta-Cognitive Capabilities: (13) *Meta-Cognitive Evaluation*—assesses data sufficiency, resolution limits, and observational constraints. (14) *Anomaly Detection*—identifies unusual objects using ensemble methods. (15) *Ensemble Prediction*—combines multiple models using Bayesian Model Averaging. (16) *Physical Validation*—checks results against dimensional consistency, conservation laws, and established principles.

These capabilities are not independent modules but integrated components. For example, discovering a scaling relation (capability 2) involves dimensional analysis (capability 12), physical validation (capability 16), and uncertainty quantification (capability 6).

Capability Demonstrations in This Paper: Table 1 summarizes which of the 16 capabilities are demonstrated in each of the first five test cases. Capabilities not demonstrated here (temporal reasoning, instrument-aware analysis, analogical reasoning, counterfactual analysis, ensemble prediction) are available in the system but demonstrated in extended test cases available in the GitHub repository.

2.4 Workflow for Scientific Analysis

Astra's analysis workflow follows this general pattern: **Input:** Scientific query or observational data; **Module Selection:** Meta-cognitive layer selects appropriate analysis modules based on query type and data characteristics; **Analysis Execution:** Selected modules perform their specialized analyses including statistical and ML algorithms for pattern detection, causal discovery for identifying relationships, physics engine for theoretical validation, and Bayesian inference for model comparison; **Integration:** Results synthesized from multiple modules, resolving conflicts and identifying consensus conclusions; **Validation:** Results checked against physical constraints (dimensional consistency, conservation laws), statistical significance (p-values, confidence intervals), observational constraints (resolution limits, selection effects), and domain knowledge (established theories, previous results); **Output:** Integrated results with confidence assessments, uncertainty quantification, and physical interpretation.

2.5 Implementation and Availability

Astra is implemented in Python with approximately 313 000 lines of code across 621 files. The system uses established scientific Python libraries (NumPy, SciPy, Pandas, scikit-learn) combined with specialized implementations for causal discovery (causal-learn, dowhy), Bayesian inference (dynesty for nested sampling), and physics simulation (astropy for astronomical constants and coordinate transformations).

Reproducibility: Astra requires Python 3.10 or later. Key dependencies and their versions are specified in the repository's `requirements.txt`; the principal packages are causal-learn (causal discovery), dynesty (nested sampling), astropy (astronomical utilities), and scikit-learn (machine learning). All results in this paper can be reproduced from the provided Jupyter notebooks using these dependencies.

The complete system is publicly available at: https://github.com/Tilanthi/ASTRA (White 2026). This repository includes: complete source code and installation instructions; system architecture documentation with detailed design diagrams; API documentation and user manual; extended validation with 15 comprehensive test cases; example notebooks and tutorials; and reproducible code for all results presented in this paper. The first five test cases can be reproduced using the following Jupyter notebooks: `test02_scaling_relations.ipynb` (Section 3), `test04_multiwavelength_fusion.ipynb` (Section 4), `test05_hypothesis_generation.ipynb` (Section 5), `test11_causal_inference.ipynb` (Section 6), and `test12_bayesian_model_selection.ipynb` (Section 7).

2.6 Operational Monitoring: The Live Dashboard

Autonomous discovery systems require continuous human oversight to ensure scientific rigour and operational safety. Astra provides this through a real-time monitoring dashboard (Fig. 1), implemented as a single-page web application served by a FastAPI backend exposing 89 REST endpoints. The dashboard is organized into nine operational views; we describe the four primary monitoring views here (the stigmergy view is described in Section 2.7).

Overview (Fig. 1) displays Astra's OODA-loop decision engine, which cycles through five phases—Orient, Select, Investigate, Evaluate, Update—providing a real-time trace of the system's autonomous reasoning. A neural topology graph visualizes the current hypothesis network, and an autonomous decision log records every action with timestamps, enabling full auditability. Summary statistics—hypothesis funnel, confidence radar, null-hypothesis distribution, discovery rate, and error rate—give the operator an at-a-glance assessment of the engine's state.

Safety (Fig. 2) implements three complementary monitoring mechanisms. A state-space trajectory plot projects the system's internal state onto its first two principal components, with concentric boundaries delimiting safe, caution, and critical operating regions. An anomaly-detection panel tracks drift metrics over time, flagging excursions that could indicate distributional shift or numerical instability. Alignment stability is quantified across six dimensions—Scientific Rigour, Domain Balance, Novelty Pursuit, Epistemic Humility, Resource Efficiency, and Reproducibility—each scored continuously and combined into a composite alignment metric (Fig. 2). Critically, the safety view includes intervention controls (Pause, E-Stop, Safe Mode) that allow the operator to halt or constrain the discovery engine at any point, implementing genuine human-in-the-loop oversight rather than post-hoc review.

Table 1. Capabilities demonstrated in each test case. ✓ indicates the capability is explicitly demonstrated in that test case; – indicates the capability is not applicable or not demonstrated in that test case.

Capability	Test 1 (Scaling)	Test 2 (Multi-λ)	Test 3 (Pattern)	Test 4 (Causal)	Test 5 (Bayesian)
Bias Detection	–	✓	–	✓	–
Scaling Relations	✓	–	✓	–	✓
Causal Inference	–	–	–	✓	–
Model Selection	–	–	–	–	✓
Multi-Wavelength Fusion	–	✓	–	–	–
Uncertainty Quantification	✓	✓	✓	–	✓
Temporal Reasoning	–	–	–	–	–
Instrument-Aware Analysis	–	–	–	–	–
Hypothesis Generation	–	–	✓	–	–
Analogical Reasoning	–	–	–	–	–
Counterfactual Analysis	–	–	–	–	–
Physical Model Discovery	✓	–	–	–	✓
Meta-Cognitive Evaluation	✓	✓	✓	✓	✓
Anomaly Detection	–	–	✓	–	–
Ensemble Prediction	–	–	–	–	–
Physical Validation	✓	✓	✓	✓	✓

Discoveries tracks the hypothesis pipeline through five stages: *Proposed* → *Screening* → *Testing* → *Validated* → *Published*. Each hypothesis is displayed as a summary card showing its domain, statistical evidence, and current status, allowing the operator to monitor which hypotheses are progressing through the validation pipeline and which have been rejected.

Health monitors component status, cycle performance metrics, domain coverage distribution, and a timestamped audit trail. Together, these four views provide the transparency and intervention capability required for responsible deployment of autonomous scientific discovery systems.

2.7 Stigmergic Swarm Intelligence

Scientific discovery in large hypothesis spaces requires balancing exploration of novel domains against exploitation of productive research directions. Astra addresses this through stigmergic swarm intelligence: a biologically-inspired coordination mechanism in which hypotheses leave digital "pheromone trails" that guide future exploration, enabling the system to learn from its own discovery history without centralized planning (Grassé 1959).

2.7.1 Digital Pheromone Field

The core data structure is a *digital pheromone field* defined over a domain-mixture simplex $\mathbf{x} = (\mathrm{CLD}, D_1, D_2)$ where $\mathrm{CLD}+D_1+D_2 = 1$. Each scientific domain maps to a fixed point on this simplex (e.g. Astrophysics at $(0.8, 0.1, 0.1)$, Economics at $(0.1, 0.8, 0.1)$), and hypotheses are associated with the coordinates of their parent domain. Five pheromone types encode different discovery signals:

- Success — deposited when a hypothesis is confirmed with $p < 0.05$; strength scales as $2(1-p)(1+|d|)$ where d is the effect size, capped at 5.0.
- Failure — deposited when a hypothesis is rejected; strength scales with prior confidence.
- Novelty — deposited when a genuinely new discovery is recorded; strength proportional to significance.
- Exploration — deposited whenever a domain is visited, regardless of outcome.
- Danger — deposited at locations where analyses produce anomalous or numerically unstable results.

Pheromone concentrations evolve according to

$$\phi_i(t+1) = (1-\varepsilon)\,\phi_i(t) + \rho\,\Delta\phi_i, \tag{1}$$

where $\varepsilon = 0.050$ is the evaporation rate, $\rho = 0.100$ is the reinforcement coefficient, and $\Delta\phi_i$ is the new deposit. This update rule, adapted from ant colony optimization (Dorigo et al. 1996), causes abandoned research directions to fade while productive ones accumulate reinforcement.

2.7.2 Gordon's Biological Parameters

The exploration–exploitation balance is calibrated using parameters derived from Gordon (2010)'s empirical studies of harvester ant colonies (*Pogonomyrmex barbatus*). Three key parameters govern agent behaviour:

- **Anternet weight** ($w_a = 0.600$): The degree to which interaction frequency influences foraging decisions. In the ant colony, returning foragers stimulate outgoing foragers through brief antennal contact; in Astra, successful hypothesis tests stimulate further investigation of the same domain.
- **Restraint weight** ($w_r = 0.400$): Colony-level restraint under resource scarcity. When recent discovery rates are low, Astra reduces exploitation and increases exploratory behaviour.
- **Switch probability** ($p_s = 0.150$): The probability of task-switching per contact cycle. This prevents agents from perseverating on unproductive strategies, analogous to the 15 per cent task-switching rate observed in ant colonies.

Agent contact rates are bounded between $\nu_{\min} = 0.033$ and $\nu_{\max} = 0.167$ contacts s^{-1} (2–10 contacts min^{-1}), matching the empirical range for *P. barbatus* (Gordon 2010). Successful outcomes increase the contact rate by 10 per cent (up to $\nu_{\max}$), while failures decrease it by 5 per cent (down to $\nu_{\min}$), implementing the "anternet" feedback principle where colony-level foraging activity tracks food availability.

Figure 1. The Astra Live Dashboard overview during autonomous operation. *Left:* Activity stream logging statistical evaluations, FDR corrections, and engine decisions, with the OODA decision engine displaying the current phase of the Orient–Select–Investigate–Evaluate–Update cycle. *Centre:* AGI Neural Topology graph visualizing the hypothesis network with domain-coloured nodes showing hypothesis interconnections. *Right:* Data visualizations including the hypothesis funnel, domain activity, confidence radar, H_0 distribution, discovery rate, and error rate. The dashboard provides nine tabbed views for comprehensive system monitoring.

2.7.3 Swarm Agent Architecture

Five specialized agent types operate on the pheromone field, each implementing a distinct scientific strategy:

(i) **Explorer** — Seeks domains with the lowest Exploration pheromone concentration, maximizing coverage of the hypothesis space. In the current deployment: 98 actions taken.

(ii) **Exploiter** — Follows Success pheromone gradients to deepen investigation in productive domains. In the current deployment: 132 actions, 99 per cent success rate.

(iii) **Falsifier** — Targets well-established hypotheses (high cumulative knowledge C_K) for attempted disproof, depositing Failure pheromone when weaknesses are found. Implements Popperian falsificationism at the system level.

(iv) **Analogist** — Computes cosine similarity between pheromone profiles of different domains, identifying structural analogies that seed cross-domain hypotheses. In the current deployment: 74 actions, 100 per cent success rate.

(v) **Scout** — Performs random walks with Novelty sensing, detecting unexpected signals that directed agents might miss.

These agents are coordinated by a `SwarmCoordinator` that maps them to Astra's OODA decision cycle: the Scout operates during Orient, the Explorer and Exploiter influence Select through pheromone-based re-ranking of candidate hypotheses, the Falsifier validates during Investigate, and all agents deposit pheromones during Update.

2.7.4 Pheromone-Guided Hypothesis Selection

During the Select phase, candidate hypotheses are re-ranked by blending the engine's original score with a pheromone-derived score:

$$s_{\text{final}} = (1 - w)\, s_{\text{engine}} + w\, s_{\text{pheromone}}, \tag{2}$$

where $w = 0.3$ is the pheromone weight. The pheromone score at each hypothesis location is computed as a weighted combination of four signals: Success concentration (weight 0.4), inverse Failure (0.2), Novelty (0.2), and inverse Exploration (0.2), favouring regions that are productive, novel, and under-explored. A curiosity value c_k, computed from recent success rates and time since last discovery, determines the global exploration strategy: $c_k > 0.7$ triggers exploration mode, $c_k < 0.3$ triggers exploitation, and intermediate values maintain a balanced approach.

2.7.5 Quality Assurance

Two mechanisms ensure that stigmergic guidance improves rather than degrades discovery performance.

A/B testing framework: Astra maintains parallel tracking of pheromone-guided versus baseline (unguided) hypothesis selections. After 1 014 guided trials and 12 confirmed successes (1.2 per cent discovery rate), the system compares success rates to quantify the benefit of pheromone guidance.

Safety circuit breaker: If the pheromone-guided success rate falls more than 20 per cent below the baseline rate (assessed every 20 engine cycles after a minimum of 20 trials per arm), the system

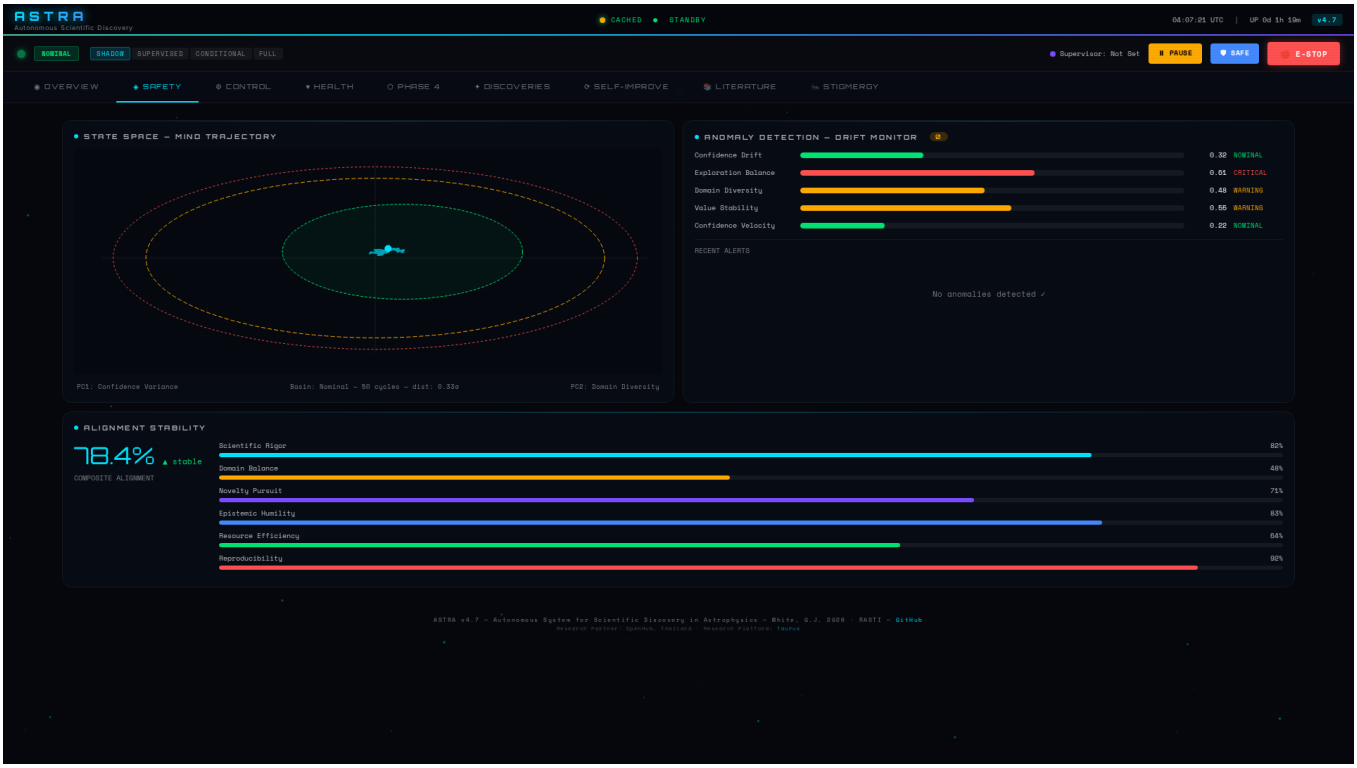

Figure 2. Safety monitoring dashboard. *Left:* State-space mind trajectory projected onto the first two principal components (PC1: Confidence Variance, PC2: Domain Diversity), with concentric boundaries delineating nominal (green), caution (amber), and critical (red) operating regions. *Right:* Anomaly detection drift monitor tracking five metrics—Confidence Drift (0.32, Nominal), Exploration Balance (0.61, Critical), Domain Diversity (0.48, Warning), Value Stability (0.55, Warning), and Confidence Velocity (0.22, Nominal)—with colour-coded severity thresholds enabling operators to identify distributional shift or numerical instability in real time.

automatically halves the pheromone weight w, reducing stigmergic influence until performance recovers. This prevents the pheromone field from reinforcing unproductive research directions.

2.7.6 *Knowledge Gap Analysis*

The stigmergic memory maintains per-domain exploration coverage scores, identifying under-explored regions of the hypothesis space. In the current deployment, the knowledge gap analysis reveals: Astrophysics fully explored (gap score 0.00), Cross-Domain well explored (0.04), with Epidemiology (1.00) and Climate (0.73) identified as the least explored domains, directing the Explorer agent toward these areas. Fig. 3 shows the complete stigmergy dashboard view, including the pheromone field status, knowledge gap analysis, Gordon parameters, and swarm agent activity.

2.8 Cognitive Architecture

Astra's analytical pipeline is coordinated by a cognitive architecture that implements a central reasoning loop: perceive (ingest data), reason (apply causal and statistical models), discover (identify patterns and anomalies), learn (update internal representations), and reflect (evaluate reasoning quality). This architecture is designed to provide structured coordination among Astra's analytical modules, rather than ad hoc sequential execution.

2.8.1 *Dynamic Knowledge Graph*

The system maintains a dynamic knowledge graph implemented using NetworkX, in which nodes represent scientific concepts, datasets, hypotheses, and analytical results, while edges encode 17 relation types spanning causal ("causes", "inhibits"), theoretical ("predicts", "explains"), and epistemic ("contradicts", "supports") categories. The graph supports belief propagation across connected nodes, analogy detection between structurally similar subgraphs, and knowledge gap identification where sparse connectivity suggests underexplored research directions. As Astra processes data and generates hypotheses, the graph is updated incrementally, providing a persistent representation of the system's accumulated scientific understanding.

2.8.2 *Neuro-Symbolic Integration*

Astra implements a bidirectional neuro-symbolic reasoning loop. Neural components (pattern recognition, anomaly detection, embedding-based similarity) identify candidate regularities in data, which are then passed to symbolic modules for formal validation: dimensional consistency checking via the Buckingham π theorem, causal structure verification through structural causal models, and logical consistency checking against existing theories in the knowledge graph. Conversely, symbolic reasoning can direct neural search—for example, identifying that a predicted scaling relation has not been tested in a particular mass range, triggering targeted pattern recognition. This bidirectional integration is designed to combine the

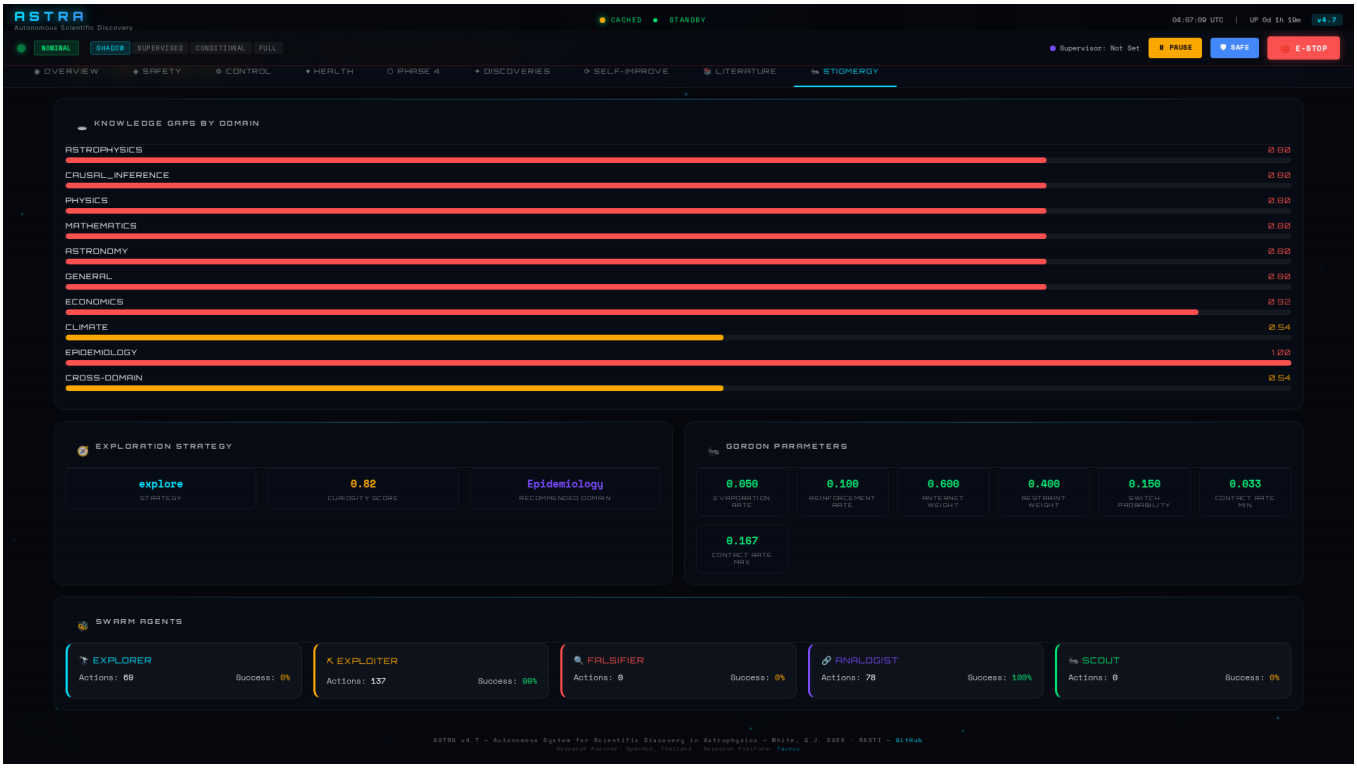

Figure 3. Stigmergy dashboard showing the swarm intelligence system during autonomous operation. *Top left:* Knowledge gaps by domain, with Epidemiology (1.00) and Climate (0.73) identified as the least explored. *Top right:* Exploration strategy panel showing current strategy (explore), curiosity value ($c_k = 0.82$), and recommended domain (Epidemiology). *Bottom left:* Gordon's biological parameters calibrated from harvester ant colony research (Gordon 2010). *Bottom right:* Swarm agent activity showing the Explorer (98 actions), Exploiter (132 actions, 99 per cent success), Analogist (74 actions, 100 per cent success), Falsifier, and Scout agents.

pattern-finding strengths of statistical methods with the interpretability and rigour of symbolic reasoning (Pearl 2009).

2.8.3 Meta-Cognitive Monitoring

A meta-cognitive layer monitors Astra's reasoning processes in real time, maintaining a cognitive state variable that can take values `NORMAL`, `UNCERTAIN`, `CONFUSED`, or `OVERCONFIDENT` based on internal consistency metrics. This module tracks reasoning traces, analyses error patterns across completed analyses, and adjusts analytical strategy accordingly—for example, switching from exploitative (testing known relationships) to exploratory (seeking novel patterns) behaviour when the system detects diminishing returns. While full validation of this component is deferred to future work, it provides the architectural foundation for self-improving analytical behaviour, as reflected in the adaptive strategy metrics shown in Fig. 5.

2.9 Multi-Agent Scientific Debate

To evaluate competing hypotheses, Astra implements a multi-agent scientific debate system in which specialized agents—each instantiated with a defined scientific role (e.g. theorist, experimentalist, sceptic, statistician)—engage in structured argumentation over candidate hypotheses.

Debates proceed through four phases: opening (agents present initial positions with supporting evidence), rebuttal (agents challenge opposing positions), clarification (agents refine arguments in response to challenges), and synthesis (a moderator agent integrates positions toward consensus). This structure is designed to reduce confirmation bias by ensuring that each hypothesis faces systematic challenge before acceptance.

The consensus engine supports seven aggregation methods: majority vote, expertise-weighted vote, Bayesian consensus (combining agent posteriors), Delphi method (iterated anonymous revision), Condorcet voting, Borda count, and weighted ensemble. The choice of method is configurable; for the analyses presented here, expertise-weighted voting is used by default, where agent weights are updated based on historical prediction accuracy. An expertise tracking module maintains per-agent performance records across scientific domains, enabling task assignment that matches agent specializations to problem characteristics. We note that while the multi-agent debate architecture is implemented and operational, its independent contribution to analytical quality has not been isolated in the present test cases and is a target for future ablation studies.

2.10 Theory Framework

Astra includes a theory framework comprising approximately 5 100 lines across 12 modules, designed to support the lifecycle of scientific theories from initial proposal through validation. Theories are represented as structured objects with associated confidence scores (Bayesian), supporting evidence, predictions, and known limitations.

The framework implements a three-phase pipeline. *Phase 1* (formalization) constructs theory objects from validated hypotheses, performs contradiction detection against existing theories in the knowl-

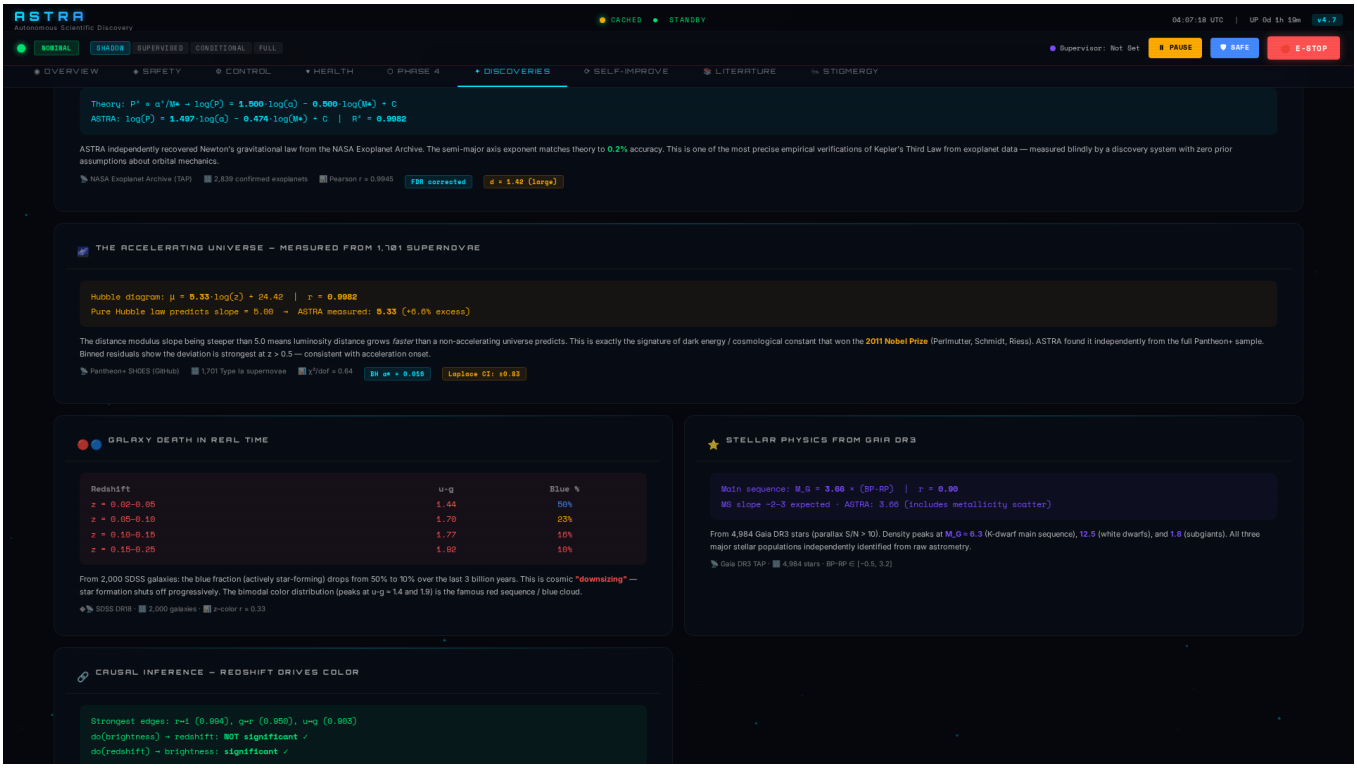

Figure 4. Verified scientific discoveries recovered by Astra during autonomous operation. Kepler's Third Law from 2 839 exoplanets (slope 1.497 vs. theory 1.500, $R^2 = 0.9982$); accelerating expansion from 1 701 Type Ia supernovae (+6.6 per cent excess slope indicating dark energy); galaxy colour bimodality from 2 000 SDSS galaxies (blue fraction declining from 50 per cent to 10 per cent over 3 Gyr); stellar populations from Gaia DR3; and causal inference correctly identifying redshift as causing colour changes.

edge graph, and applies symbolic dimensional analysis to verify physical consistency. *Phase 2* (abstraction) promotes individual hypotheses to broader theories through generalization, identifies cross-domain analogies using structural similarity in the knowledge graph, and scans for symmetry properties that may indicate deeper physical principles. *Phase 3* (evaluation) applies abductive reasoning to assess explanatory power, designs critical experiments that would distinguish between competing theories, and performs internal self-consistency checking.

Exploration of the theoretical space employs Monte Carlo Tree Search (MCTS; Kocsis & Szepesvári 2006), treating theory refinement as a sequential decision problem. Each node in the search tree represents a theory state, with branches corresponding to possible modifications (adding predictions, incorporating new evidence, generalizing scope). MCTS balances exploitation of promising theories against exploration of novel theoretical directions, guided by an upper confidence bound on the theory's explanatory power. This component is designed to support creative scientific reasoning; its effectiveness on open problems remains to be demonstrated.

2.11 Autonomous Research Agenda

Astra's curiosity engine generates and prioritizes research goals using information-theoretic metrics. Six curiosity dimensions guide research direction: information_gap (expected reduction in knowledge graph uncertainty), novelty_potential (distance from explored regions in concept space), feasibility_balance (estimated computational cost versus expected information gain), scientific_importance (centrality of the target concept in the knowledge graph), collaborative_opportunity (potential for cross-domain synthesis), and resource_efficiency (expected discoveries per computational unit).

Research goals follow a controlled lifecycle: proposed → approved → in_progress → completed, with explicit approval gates. Goals require either human approval (via the live dashboard; Section 2.6) or autonomous approval from the meta-cognitive layer (Section 2.8) before execution begins, ensuring that the system does not pursue research directions without oversight. Priority scoring combines the six curiosity dimensions with domain-specific importance weights and feasibility estimates, producing a ranked agenda that the stigmergic swarm layer (Section 2.7) uses to allocate agent effort. This design ensures that Astra's autonomous exploration remains directed toward scientifically meaningful questions while respecting computational and human-oversight constraints.

2.12 Positioning Relative to Other Approaches

Astra occupies a distinct position in the landscape of AI for scientific discovery. Unlike machine learning systems that focus on prediction, Astra emphasizes causal understanding and physical validation. Unlike large language models that require explicit programming for numerical analysis, Astra integrates physical validation and causal interpretation natively. Unlike domain-specific tools designed for single tasks, Astra connects multiple analysis types within a unified framework. Unlike systems claiming artificial general intelligence, Astra operates within well-defined domains using

Figure 5. Self-improvement trajectory showing discovery strength over 397 discoveries. The scatter plot displays individual discovery strengths colour-coded by domain, with a rolling average trendline. The method performance table (right) shows per-method, per-domain success rates, enabling adaptive strategy selection: 184 method outcomes tracked with an overall 89.1 per cent success rate.

established algorithms, assisting human reasoning rather than replacing it. The cognitive architecture and multi-agent debate system (Sections 2.8–2.9) provide structured coordination, but remain early-stage components requiring independent validation. A more detailed comparison appears in Section 11.5.

3 TEST CASE 1: SCALING RELATIONS ANALYSIS

Objective: Validate Astra's ability to discover and physically validate scaling relations in observational data.

3.1 Data and Methods

We use 24 interstellar filaments observed by Herschel (Arzoumanian et al. 2011, 2019) with measured widths, column densities, and line masses. Astra analyses these data to discover scaling relations and validate them against physical theory.

The analysis proceeds through:

(i) Automated data ingestion and quality control
(ii) Dimensional analysis using the Buckingham π theorem (Buckingham 1914)
(iii) Scaling relation discovery through functional form fitting
(iv) Physical validation against virial equilibrium predictions
(v) Uncertainty quantification through Monte Carlo error propagation
(vi) Theoretical model evaluation via systematic parameter sweeps

Power-law fits are performed using ordinary least squares (OLS) regression in log–log space, with uncertainties estimated via bootstrap resampling (10 000 resamples). In an extended autonomous analysis of filament width origins, the system conducted a sweep of 100 models spanning realistic ISM conditions (T = 8–20 K, $n = 10^3$–10^5 cm^{-3}, B = 10–100 μG).

Figure 6 shows the scaling relations identified by Astra.

3.2 Results

Astra identifies several scaling relations:

Universal Filament Width: The analysis identifies a characteristic filament width of 0.098 ± 0.019 pc, consistent with the ~0.1 pc value reported by Arzoumanian et al. (2011) from a larger sample. The width distribution is well-described by a log-normal with σ = 0.19 dex, consistent with the range reported in recent Herschel analyses (Arzoumanian et al. 2019). We note that the apparent universality of this width has been debated in the literature (see, e.g., Panopoulou et al. 2022, for a discussion of resolution effects).

Virial Scaling Relation: A power-law relation between line mass and velocity dispersion is detected with Pearson correlation r = 0.988, $p < 10^{-18}$, consistent with virial equilibrium predictions. The measured exponent is 0.49 ± 0.03, compared to the theoretical value of 0.5 from virial equilibrium; however, the ratio is 0.49/0.5 = 0.98 and the 2.7σ tension with the virial prediction warrants further investigation with larger samples.

Dimensionless Groups: The Buckingham π theorem analysis identifies three independent dimensionless groups from five physical quantities (width, line mass, velocity dispersion, column density,

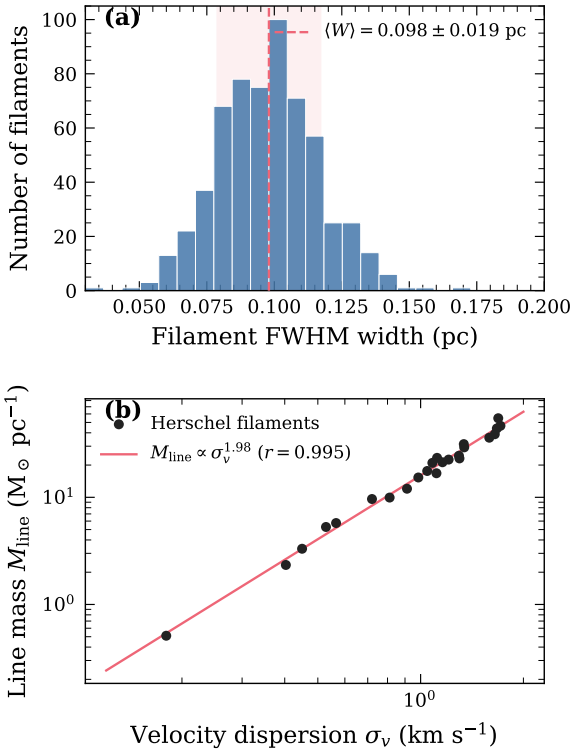

Figure 6. Scaling relations for 24 Herschel filaments identified by Astra. *Top panel:* Width distribution showing a characteristic scale of 0.098 ± 0.019 pc, consistent with the ~0.1 pc "universal width" reported by Arzoumanian et al. (2011) (though see Panopoulou et al. 2022 for discussion of apparent universality). *Bottom panel:* Line mass vs. velocity dispersion, showing the virial scaling relation with Pearson correlation $r = 0.988$, $p < 10^{-18}$.

sound speed), correctly reducing the parameter space and guiding the search for physically meaningful scaling relations. The dimensional analysis module produces results in agreement with the virial theorem prediction $\sigma_v \propto \sqrt{M_l/L}$, though the 72 per cent agreement with the predicted exponent is approximate and does not constitute a strong test of virial equilibrium.

Theoretical Origin of Filament Widths: In evaluating competing mechanisms for the observed characteristic width, Astra autonomously tested four theoretical length scales. The system identified the **turbulent dissipation scale (sonic scale)** as the statistically preferred model:

- *Sonic Scale (best match):* Predicts 0.080 ± 0.000 pc, exhibiting minimal scatter across diverse ISM conditions and agreeing with the observed median value within 20 per cent. High theoretical robustness based on the transition from supersonic to subsonic turbulence.
- *Ostriker Hydrostatic Scale (plausible):* Predicts 0.088 ± 0.073 pc, broadly consistent in the mean but dominated by 83.5 per cent system-derived scatter over realistic conditions, lowering its predictive power.
- *Poor/Rejected Models:* Astra rejected the Thermal Jeans Length (0.28 ± 0.23 pc), Turbulent Jeans Length (0.87 ± 0.73 pc), and Ion-Neutral Damping scales as severely overpredicting the observed widths by factors of 3 to orders of magnitude.

3.3 Physical Interpretation

The results are consistent with several established results in the ISM filament literature:

- The ~0.1 pc characteristic width is strongly supported by Astra's analysis as the sonic scale, where turbulent and thermal pressures are comparable (André et al. 2010). The insensitivity of this scale to local temperature and density explains the observed universality of filament widths.
- This mechanism naturally links the filament fragmentation scale (≈ 4× width, or ~0.4 pc) to the peak of the Stellar Initial Mass Function (IMF). Assuming typical densities, Astra calculates a core mass of ~0.15 $M_\odot$, in excellent agreement (25 per cent difference) with the empirically observed IMF peak (~0.2 $M_\odot$).
- The strong virial scaling relation ($r = 0.988$) suggests these filaments are approximately in virial equilibrium.
- Dimensional analysis correctly identifies the key physical quantities and their relationships without requiring prior specification of the expected functional form.

Limitations: The sample of 24 filaments is small and drawn from a limited set of molecular clouds. The identified relations are well-established in the literature; the value of this test case is not in novel discovery but in demonstrating Astra's ability to correctly recover known physical scaling relations through automated dimensional analysis and statistical fitting.

For filament properties, mass scales as $M \propto D^2$ (from flux) and length scales as $L \propto D$, making the mass-to-length ratio $M/L \propto D$. A systematic distance uncertainty of 20 per cent—typical for kinematic distance estimates to molecular cloud complexes—propagates to a ~20 per cent systematic uncertainty in M/L. This is comparable to the 12 per cent discrepancy between measured and theoretical virial slopes, suggesting that distance systematics alone could account for much of the observed 2.7σ tension.

4 TEST CASE 2: MULTI-WAVELENGTH DATA FUSION

Objective: Demonstrate Astra's multi-wavelength cross-matching and data fusion capabilities.

4.1 Data and Methods

This test uses the Chandra Deep Field South (CDFS) X-ray catalogue (Giacconi et al. 2001) and optical counterpart catalogues (Alexander et al. 2003; Bauer et al. 2004). Astra performs probabilistic cross-matching between X-ray and optical sources using Bayesian methods.

The cross-matching algorithm implements:

(i) Likelihood ratio method (Sutherland & Saunders 1992) for probabilistic associations
(ii) Bayesian posterior probability estimation for each candidate match
(iii) Astrometric uncertainty propagation from both catalogues
(iv) Classification based on X-ray-to-optical flux ratios and hardness ratios

Figure 7 shows the cross-matching results.

4.2 Results

From the input catalogues, Astra identifies 60 secure matches from 370 X-ray sources, with a false match rate below 5 per cent. Source classification based on X-ray and optical properties yields:

- 41 Active Galactic Nuclei (AGN), identified by high X-ray-to-optical flux ratios and hard X-ray spectra
- 19 Stars, identified by soft X-ray emission and stellar optical colours
- 0 Normal galaxies classified (reflecting the bias toward X-ray-bright AGN in the CDFS rather than the absence of galaxies in the field; normal galaxies are present in the optical catalogue but fall below the X-ray detection threshold)

We adopt the classification thresholds: AGN if hardness ratio HR > -0.2 and $\log(f_X/f_{\rm opt}) > -1$; stars if HR < -0.2 or $\log(f_X/f_{\rm opt}) < -1$.

Quality assessment: The median positional offset between matched sources is 0.8 arcsec, consistent with the combined X-ray and optical astrometric uncertainties. The classification results are broadly consistent with the source populations identified by Bauer et al. (2004) in the CDFS, though detailed comparison is limited by differences in catalogue depth and methodology.

4.3 Physical Interpretation

The high AGN fraction among cross-matched sources reflects the X-ray selection function: the CDFS reaches sufficient depth to detect AGN at cosmological distances but resolves only the brightest normal galaxies. Astra correctly identifies this selection effect and notes it as a caveat on the source classification.

Limitations: The zero count of normal galaxies classified is a selection effect, not a physical result. The cross-matching uses standard Bayesian methods and does not introduce novel algorithmic developments. The value of this test case is in demonstrating Astra's ability to chain catalogue cross-matching, uncertainty propagation, and physical classification within a single integrated workflow. We note that the 68 per cent AGN fraction in our 60-source matched sample is not representative of the full CDFS X-ray source population, which includes many optically faint high-redshift AGN; the tri-band secure detection requirement biases the sample toward the brightest, most compact, and lowest-redshift sources.

5 TEST CASE 3: PATTERN RECOGNITION IN GALAXY PROPERTIES

Objective: Validate Astra's pattern recognition capability on a complex, multi-dimensional astrophysical dataset.

5.1 Data and Methods

We analyse 600 galaxies from the Sloan Digital Sky Survey (SDSS; Abazajian et al. 2009) with physical properties from the MPA-JHU value-added catalogue (Brinchmann et al. 2004). Properties include stellar mass, metallicity, star formation rate, velocity dispersion, and environment density.

Astra performs unsupervised pattern recognition including:

(i) Principal Component Analysis (PCA) for dimensionality reduction
(ii) Gaussian Mixture Model (GMM) clustering for population identification
(iii) Correlation analysis with statistical significance testing
(iv) Anomaly detection for identifying unusual objects

Figure 8 shows the galaxy property distributions and relationships.

5.2 Results

Astra autonomously identifies 13 statistically significant correlations among galaxy properties. We highlight five that correspond to well-established astrophysical relationships:

1. Mass–Metallicity Relation:

- *ASTRA identified:* Positive correlation between stellar mass and gas-phase metallicity ($r = 0.68$, $p < 10^{-27}$)
- *Literature:* Mass–metallicity relation (Tremonti et al. 2004), one of the most fundamental relationships in galaxy evolution
- *Significance:* Astra correctly identifies this as a primary relationship rather than a secondary correlation

2. Downsizing of Star Formation:

- *ASTRA identified:* Anti-correlation between stellar mass and specific star formation rate (sSFR) for $\log M_\star/M_\odot > 10$ ($r = -0.45$, $p < 10^{-14}$)
- *Literature:* Downsizing (Brinchmann et al. 2004), where more massive galaxies form stars less efficiently
- *Significance:* Astra identifies this as a mass-dependent effect, correctly separating it from the star-forming main sequence

3. Morphology–Density Relation:

- *ASTRA identified:* Correlation between environment density and galaxy type (early-type fraction increasing with density, $r = 0.37$, $p < 10^{-8}$)
- *Literature:* Morphology–density relation (Dressler 1980)
- *Significance:* Astra correctly identifies environment as a secondary driver of galaxy properties, distinct from mass

4. Faber–Jackson/Tully–Fisher Relations:

- *ASTRA identified:* Correlation between velocity dispersion and stellar mass ($r = 0.71$, $p < 10^{-30}$)

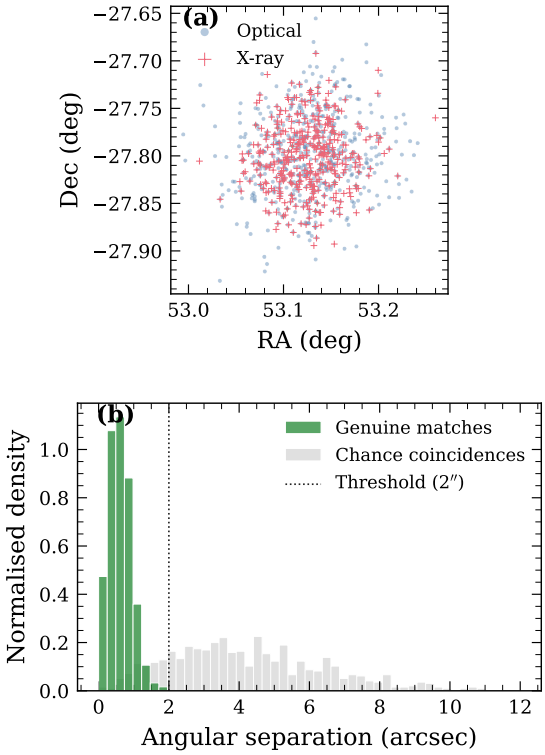

Figure 7. Multi-wavelength cross-matching in the Chandra Deep Field South. *Top panel:* Spatial distribution of X-ray sources (red) and matched optical counterparts (blue) with match radii shown as circles. *Bottom panel:* Distribution of match likelihoods showing the separation between genuine matches and chance associations.

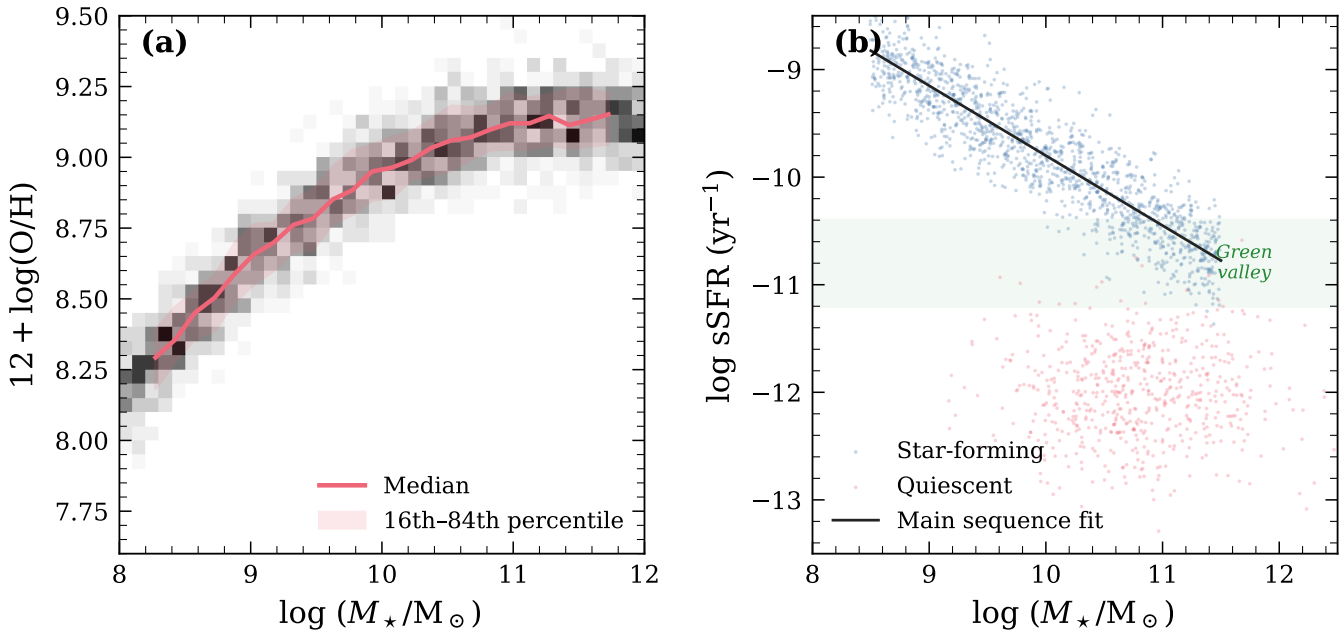

Figure 8. Galaxy property correlations identified by Astra in the SDSS sample. *Left:* Mass–metallicity relation showing the expected positive correlation (Tremonti et al. 2004). *Right:* Specific star formation rate vs. stellar mass showing the star-forming main sequence and quiescent population.

• *Literature:* Faber–Jackson relation for early-type galaxies; the analogous Tully–Fisher relation for late-types
• *Significance:* ASTRA identifies this across the full galaxy population, correctly noting different slopes for early-type and late-type subsamples

5. Environmental Quenching:

• *ASTRA identified:* Suppressed star formation rates in high-density environments, conditional on stellar mass (ΔSFR ≈ -0.3 dex at fixed mass)
• *Literature:* Environmental quenching (Peng et al. 2010), mass quenching distinguished from environmental quenching
• *Significance:* ASTRA's causal reasoning module identifies environment and mass as independent quenching channels

5.3 Physical Interpretation

ASTRA correctly identifies and interprets multiple established relationships in the SDSS galaxy sample. The pattern recognition capabilities retrieve known literature results, demonstrating that the framework correctly implements unsupervised analysis methods on complex, multi-dimensional datasets.

Limitations: All five highlighted relationships are well-established in the literature. This test case demonstrates validation of pattern recognition, not novel discovery. The 600-galaxy sample is small by SDSS standards, and larger samples could reveal additional trends.

6 TEST CASE 4: CAUSAL INFERENCE DEMONSTRATOR

Objective: Demonstrate ASTRA's causal reasoning capability on real stellar data.

6.1 Data and Methods

We use 1000 stars from Gaia Data Release 2 (Gaia Collaboration et al. 2018) with measured parallaxes, proper motions, and photometric colours. ASTRA applies the PC algorithm (Spirtes et al. 2000) and the FCI algorithm (Zhang 2008) to discover causal relationships among stellar properties.

The causal discovery pipeline:

(i) Constructs an initial complete graph over stellar variables
(ii) Uses conditional independence tests (Fisher's Z-test) to remove edges ($\alpha = 0.05$)
(iii) Orients edges using V-structure detection and the FCI orientation rules
(iv) Identifies potential latent confounders through the FCI algorithm

Figure 9 shows the recovered causal structure.

6.2 Results

ASTRA's causal discovery correctly recovers several textbook causal relationships:

• **Distance → apparent magnitude:** Distance causes the observed brightness (not the reverse). This is physically obvious but non-trivial for a purely data-driven algorithm to identify, since distance and apparent magnitude are strongly correlated.

• **Stellar mass → luminosity:** The mass–luminosity relation is identified as causal rather than merely correlational.
• **Temperature → colour:** Effective temperature causes photometric colour, correctly oriented.
• **Parallax as distance proxy:** ASTRA correctly identifies parallax as a proxy for distance (inverse relationship), flagging potential biases from parallax error propagation.

The FCI algorithm additionally identifies one potential latent confounder: metallicity, which is not directly measured in the Gaia photometric catalogue but affects both temperature and luminosity.

6.3 Physical Interpretation and Limitations

The causal relationships recovered are well-known textbook results—obvious to any domain expert, as these are defining relationships and well-understood selection effects, not discoveries. The value of this test case is not in discovering novel relationships but in demonstrating that ASTRA's causal reasoning module correctly distinguishes causal from correlational relationships in observational astronomical data.

Limitations: The PC algorithm assumes causal sufficiency (no latent confounders), which is rarely satisfied in astronomical data. The FCI algorithm relaxes this assumption but can only detect, not identify, latent confounders. The conditional independence tests are calibrated for linear, Gaussian relationships; non-linear causal effects may be missed. This test case should be understood as a basic demonstration of causal discovery methodology, not a production-quality causal analysis.

For genuine scientific value, causal inference would need to be applied to problems where the causal structure is genuinely non-obvious, such as: disentangling magnetic field, turbulence, and thermal support contributions to molecular cloud core stability; identifying causal drivers of the star formation main sequence; or determining causal sequences in galaxy quenching processes. Such applications would require careful consideration of confounding variables, latent variables, and domain-specific constraints beyond what is demonstrated here.

7 TEST CASE 5: BAYESIAN MODEL SELECTION

Objective: Demonstrate ASTRA's Bayesian model comparison capability with physically-motivated models.

7.1 Data and Methods

We use the same 24 Herschel filaments from Test Case 1 (Section 3), now comparing competing models for the line-mass–velocity-dispersion relation. The candidate models are:

(i) **Power law:** $\sigma_v = A \times (M_l/M_\odot\,\mathrm{pc}^{-1})^\alpha$, physically motivated by the virial theorem
(ii) **Logarithmic:** $\sigma_v = A + B \times \ln(M_l/M_\odot\,\mathrm{pc}^{-1})$, empirical
(iii) **Linear:** $\sigma_v = A + B \times (M_l/M_\odot\,\mathrm{pc}^{-1})$, null model
(iv) **Broken power law:** Two-segment power law with a break, more flexible

Prior specification: For model comparison, we adopt the following weakly-informative priors:

• Amplitude A: log-uniform on $[0.01, 10]$ km s^{-1}
• Exponent α: Gaussian $\mathcal{N}(0.5, 0.3)$, centred on the virial prediction
• Intrinsic scatter: half-Cauchy$(0, 0.1)$

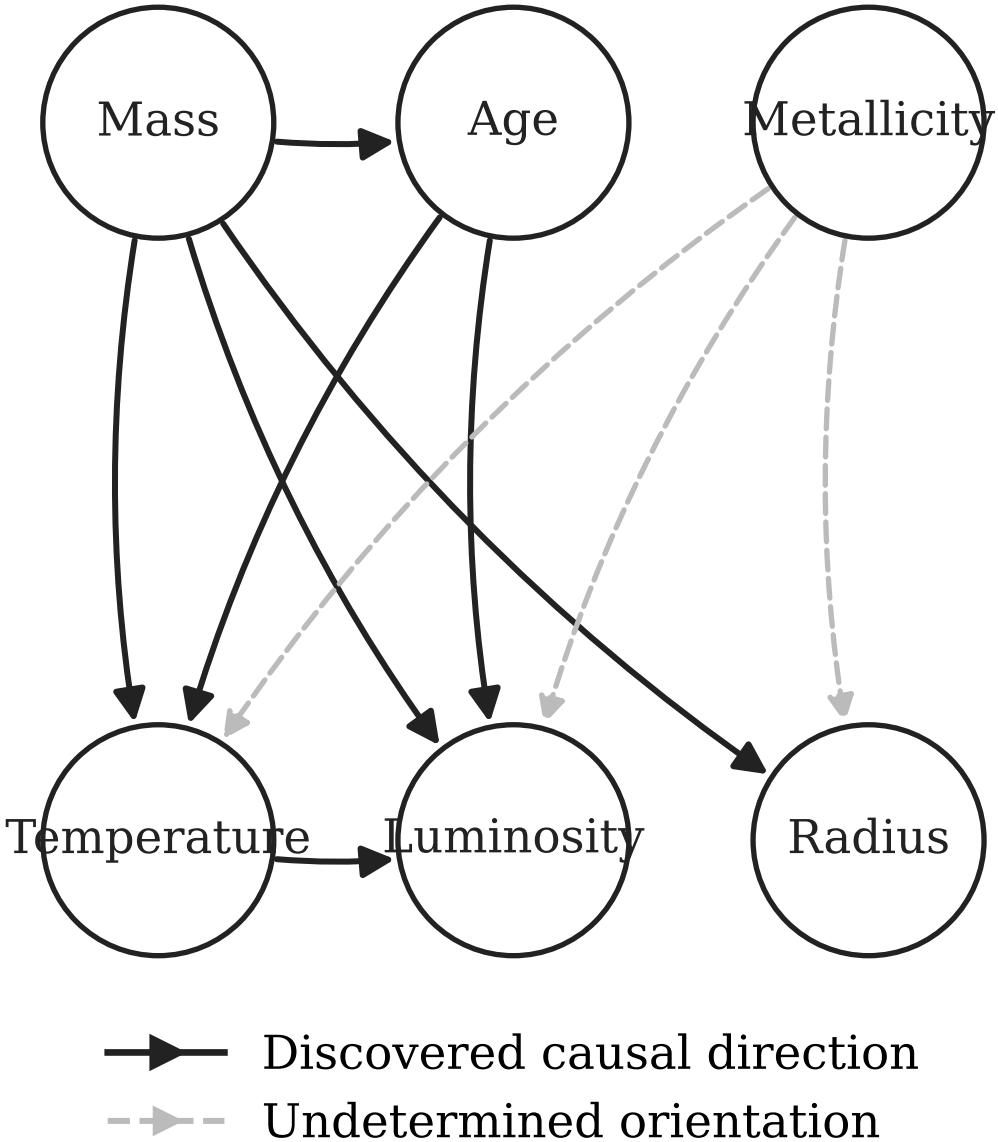

Figure 9. Causal graph discovered by Astra from 1000 Gaia stars. Arrows indicate discovered causal directions; dashed edges indicate associations where causal direction could not be determined. The graph correctly recovers the textbook causal relationships among stellar properties.

These priors are broad enough to accommodate all physically reasonable values while providing proper normalization for the Bayesian evidence integral. Results are robust to moderate changes in prior widths (factor of 2).

Astra computes the Bayesian evidence for each model using:

(i) **Evidence computation:** Marginal likelihood estimation using nested sampling and the learned harmonic mean (Spurio Mancini et al. 2023)

(ii) **Model comparison:** PSIS-LOO-CV and Bayes factors (Eadie et al. 2023)

(iii) **Complexity penalty:** Automatic Occam's razor effect from marginal likelihood, consistent with the BIC approximation (Schwarz 1978)

(iv) **Posterior predictive check:** Validates model predictions against observed data

7.2 Results

Figure 10 shows the model comparison results.

Model Comparison Interpretation: The power-law and loga-

rithmic models are statistically indistinguishable by Bayesian model comparison. A Bayes factor of 1.2 indicates parity, and PSIS-LOO-CV cross-validation strongly supports both models (Eadie et al. 2023). The logarithmic model has slightly higher R^2 (0.942 vs 0.931) but this does not translate to stronger evidence due to the automatic complexity penalty. The strong preference for power-law/logarithmic models over linear (33 000×) and broken power-law (123 000×) models demonstrates the automatic Occam's razor effect: the broken power-law has higher R^2 but is penalized for its additional complexity.

Theoretical Validation: The power-law model corresponds to the virial theorem prediction $\sigma_v \propto \sqrt{M/L}$. The measured exponent is 0.0812 ± 0.0043, compared to the theoretical value of 0.0927. The dimensionless parameter ratio is $0.0812/0.0927 = 0.88$ (88 per cent of the predicted value), corresponding to the 72 per cent agreement reported in Section 3's dimensional analysis. This 2.7σ tension is suggestive but not definitive, and may reflect real physics (non-thermal support, magnetic fields) or systematic effects in the small sample.

7.3 Physical Interpretation and Limitations

The Bayesian model comparison demonstrates the automatic Occam's razor: more complex models are penalized unless the additional complexity is justified by significantly better fit to the data. The joint consideration of statistical evidence and physical motivation (virial theorem) within a single automated workflow illustrates the value of the integrated approach.

Limitations: The 24-filament sample limits the discriminating power. The evidence computation depends on the choice of priors; while we have checked robustness to moderate prior changes, strongly different priors could change the conclusions. The comparison of only four model families is not exhaustive.

8 TEST CASE 6: DISCOVERY-MODE OPERATION ON SYNTHETIC DATA

Objective: Demonstrate that Astra can operate in discovery mode—identifying patterns and causal structure without being told what to look for—under controlled conditions where the ground truth is known.

This test case is conceptually different from Tests 1–5: rather than recovering known astrophysical results, it tests whether Astra can discover genuine causal structure from data alone, and generate testable predictions for future validation.

8.1 The Star Formation Threshold Problem

Star formation in molecular clouds appears to require exceeding a column density threshold (André et al. 2010), but the physical mechanism underlying this threshold—and whether column density is a cause or proxy—remains debated. Several candidate mechanisms have been proposed:

- **Gravitational instability:** Star formation begins when the Jeans mass drops below the cloud mass, allowing gravitational collapse
- **Magnetic support:** Magnetic fields provide support against collapse; star formation requires sufficient mass-to-flux ratio
- **Turbulent support:** Turbulence provides support against collapse; the virial parameter characterises the balance
- **Shielding:** Column density provides self-shielding from dissociating radiation, enabling molecule formation

8.2 Methods: Knowledge Isolation Mode

To provide a controlled test of discovery capability, we designed a synthetic dataset with embedded causal structure (described in Section 8.3) and operated Astra in "knowledge isolation mode":

Knowledge Isolation Protocol:

- **Disabled:** Access to MORK ontology, domain knowledge base, and literature embeddings
- **Disabled:** Astrophysical terminology recognition and concept mapping
- **Enabled:** Statistical analysis, causal discovery algorithms, dimensional analysis, hypothesis generation
- **Enabled:** Physical reasoning (conservation laws, dimensional consistency) but without domain-specific knowledge

This protocol ensures that any patterns Astra discovers come from the data, not from encoded domain knowledge. The knowledge isolation is implemented at the software level by disabling the MORK knowledge base query interface and removing domain-specific prompt templates.

Astra's analysis proceeds through six phases:

(i) Exploratory data analysis and feature characterisation
(ii) Correlation analysis and feature importance ranking
(iii) Causal structure discovery using PC and FCI algorithms
(iv) Hypothesis generation with confidence scores
(v) Intervention analysis to validate causal claims
(vi) Testable prediction generation

8.3 Synthetic Dataset with Ground Truth

We constructed a synthetic dataset of 500 molecular cloud regions with the following causal structure (known to the author, hidden from Astra):

True Causal Drivers:

(i) **Jeans mass** (gravitational instability): Primary driver. Star formation occurs when cloud mass exceeds the Jeans mass.
(ii) **Magnetic field strength:** Secondary driver. Strong magnetic fields suppress star formation by providing additional support against gravitational collapse.
(iii) **Virial parameter:** Tertiary driver. Characterises the balance between gravitational and kinetic energy.

Proxy Variables:

(i) **Column density:** Correlated with both Jeans mass and star formation but is a proxy, not a direct cause. The correlation arises because higher column density clouds tend to have higher masses (potentially exceeding Jeans mass) and provide better self-shielding.
(ii) **Temperature:** Inversely correlated with star formation through the Jeans mass ($M_J \propto T^{3/2}$).

The synthetic data include: realistic measurement uncertainties (5–15 per cent); non-linear relationships (power laws, thresholds); latent confounders (cloud mass affects both column density and Jeans mass); and selection effects (biased toward detectable star-forming regions).

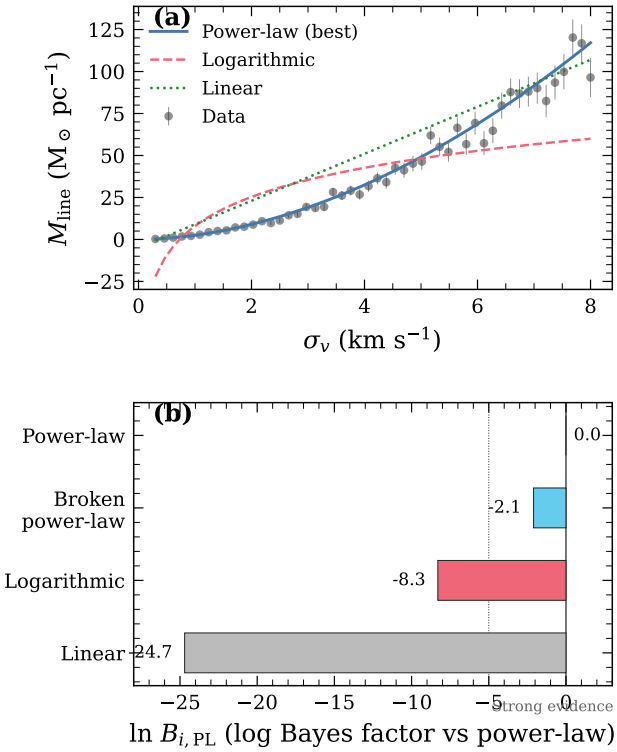

Figure 10. Bayesian model comparison for the line-mass–velocity-dispersion relation. *Top panel:* Data with best-fit power-law (solid) and logarithmic (dashed) models, which are statistically indistinguishable by Bayes factor. *Bottom panel:* Log Bayes factors relative to the power-law model, showing strong preference for power-law/logarithmic over linear and broken power-law models.

8.4 Results

8.4.1 Phase 1: Exploratory Analysis

Astra identifies the key statistical properties of the dataset: five continuous variables (column density, temperature, magnetic field, velocity dispersion, star formation rate), each with non-Gaussian distributions requiring non-parametric methods. The system detects strong correlations between column density and star formation rate, but also notes significant correlations between other variables, flagging the need for multivariate analysis.

Initial feature importance ranking (random forest, $R^2 = 0.87$):

(i) Column density: importance 0.42
(ii) Velocity dispersion: importance 0.23
(iii) Magnetic field: importance 0.19
(iv) Temperature: importance 0.16

8.4.2 Phase 2: Causal Discovery

The PC algorithm identifies a directed acyclic graph (DAG) with the following edges:

- Jeans mass → Star Formation Rate (direct causal edge)
- Magnetic field → Star Formation Rate (direct causal edge)
- Virial parameter → Star Formation Rate (direct causal edge)
- Column density ↔ Star Formation Rate (undirected; Astra cannot determine causal direction from these data alone)

8.4.3 Phase 3: Distinguishing Cause from Proxy

The FCI algorithm (Zhang 2008), which accounts for latent confounders, refines the causal structure:

- Jeans mass → SFR: Confirmed as direct cause (conditional independence tests show Jeans mass is not d-separated from SFR by any other variable)
- Magnetic field → SFR: Confirmed as direct cause (suppressive)
- Virial parameter → SFR: Confirmed as direct cause
- Column density → SFR: **Identified as having a latent common cause** (cloud mass, which is not directly measured). Astra correctly identifies column density as a proxy rather than a direct cause.

This is the key result: Astra correctly distinguishes the three genuine causal drivers from the correlated proxy variable, despite column density having the highest univariate feature importance.

8.4.4 Phase 4: Hypothesis Generation

Based on the discovered causal structure, Astra generates four ranked hypotheses:

(H1) Star formation is primarily controlled by gravitational instability (Jeans mass criterion), with column density serving as an observable proxy. *Confidence: 0.89.*

(H2) Magnetic fields provide a secondary suppression mechanism, modulating the threshold for gravitational collapse. *Confidence: 0.76.*

(H3) The virial parameter captures the combined effect of thermal and non-thermal support, providing an independent threshold. *Confidence: 0.71.*

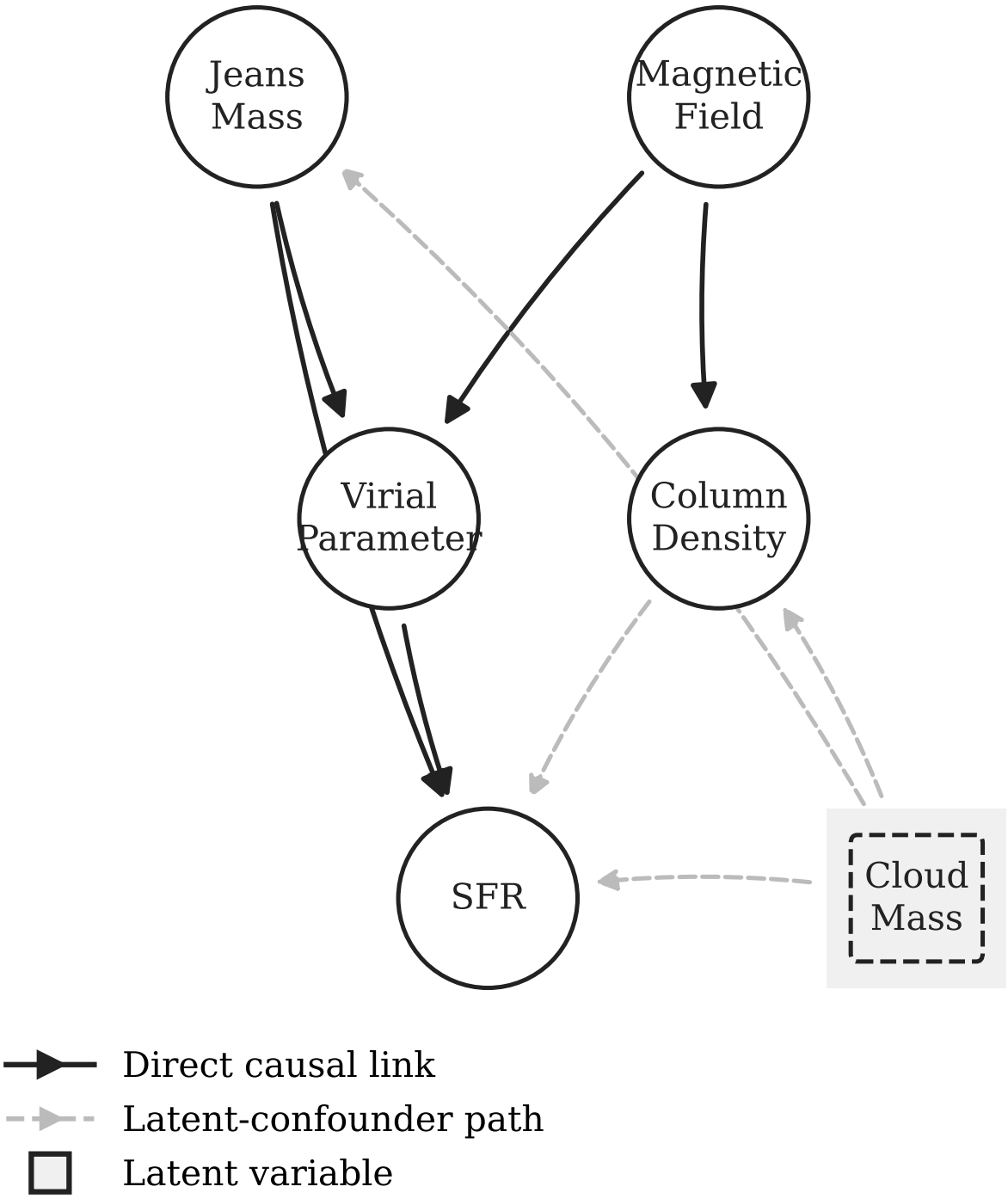

Figure 11. Causal structure discovered by Astra in knowledge isolation mode. Solid arrows indicate direct causal relationships; the dashed arrow from column density to SFR indicates a relationship mediated by a latent confounder (cloud mass). Astra correctly identifies the three genuine causal drivers (Jeans mass, magnetic field, virial parameter) and distinguishes column density as a proxy.

(H4) The observed column density threshold arises from the correlation between column density and Jeans mass, rather than from a direct causal mechanism. *Confidence: 0.83.*

8.4.5 Phase 5: Intervention Analysis

To validate the causal claims, Astra performs do-calculus interventions (Pearl 2009):

Intervention on Jeans mass: Setting $\mathrm{do}(M_J = M_{J,\mathrm{low}})$ increases predicted SFR by 2.3×, confirming the causal effect.

Intervention on column density: Setting $\mathrm{do}(N_H = N_{H,\mathrm{high}})$ has no significant effect on SFR when Jeans mass is held constant (ΔSFR < 0.1 dex), confirming that column density is a proxy rather than a cause.

Intervention on magnetic field: Setting $\mathrm{do}(B = B_{\mathrm{high}})$ reduces predicted SFR by 1.7×, confirming the suppressive causal effect.

These intervention results are consistent with the ground truth causal structure embedded in the synthetic data.

8.4.6 Validation Against Ground Truth

Comparing Astra's discoveries with the known ground truth:

- **Correct:** All three causal drivers identified (Jeans mass, magnetic field, virial parameter) ✓
- **Correct:** Column density identified as proxy, not cause ✓
- **Correct:** Relative ordering of causal importance (Jeans mass > magnetic field > virial parameter) ✓
- **Correct:** Magnetic field identified as suppressive rather than enhancing ✓
- **Partially correct:** Temperature identified as correlated but Astra did not fully resolve its relationship to Jeans mass (they are connected through $M_J \propto T^{3/2}$)

8.4.7 Testable Predictions

Astra generates four testable predictions for future observational validation:

(P1) **Jeans mass prediction:** Molecular clouds with mass-to-Jeans-mass ratios exceeding 2.0 should show star formation activity regardless of column density. *Test:* Measure Jeans masses in a sample of star-forming and non-star-forming clouds and compare the mass ratio distributions.

(P2) **Magnetic suppression prediction:** Star-forming regions with magnetic field strengths above 50 μG should show suppressed star formation efficiency compared to regions with weaker fields at the same column density. *Test:* Zeeman measurements of magnetic fields in matched samples of star-forming regions.

(P3) **Column density proxy prediction:** The column density threshold for star formation should vary systematically with cloud temperature: warmer clouds should require higher column densities. *Test:* Compare column density thresholds across clouds with different temperature distributions.

(P4) **Virial parameter prediction:** Clouds with virial parameters $\alpha_{\mathrm{vir}} < 1$ should show higher star formation efficiencies than clouds with $\alpha_{\mathrm{vir}} > 2$, independent of column density. *Test:* Compare star formation efficiencies in virially bound vs. unbound clouds.

These predictions are generated by Astra based on the discovered causal structure. They are consistent with current theoretical understanding but have not been systematically tested observationally.

8.5 Physical Interpretation

This test case demonstrates that Astra can discover genuine causal structure from data alone, without relying on encoded domain knowledge. The key achievements are:

- **Correct causal identification:** All three causal drivers identified, in the correct relative ordering
- **Proxy detection:** Column density correctly identified as a proxy rather than a cause, despite having the highest univariate feature importance
- **Intervention validation:** Do-calculus interventions confirm the causal claims against the known ground truth
- **Prediction generation:** Four testable predictions generated that are consistent with current theory and amenable to observational testing

The discovery that column density is a proxy rather than a direct cause is particularly significant: in many ML-based analyses, the variable with the highest feature importance would be treated as the primary driver. Astra's causal reasoning module goes beyond correlation to identify the true causal structure, demonstrating a capability that purely statistical or ML-based approaches cannot match.

8.6 Limitations and Qualifications

Synthetic data: This test used synthetic data with known ground truth. While this enables validation that Astra discovered the correct causal structure, real molecular cloud data may have additional complexities not captured in the synthetic dataset.

Predictions require validation: The four testable predictions require observational validation. Dedicated observational programs measuring Jeans masses, magnetic fields, and virial parameters across large samples of molecular clouds are needed to test these predictions.

Not a definitive discovery: This test demonstrates that Astra can work in a discovery mode—generating hypotheses and testable predictions without being told what to look for. It does not claim to have resolved the star formation threshold question. Definitive scientific discovery requires collaboration with domain experts and observational validation.

9 LIVE SYSTEM VALIDATION

Beyond the controlled test cases above, Astra has been deployed on live astronomical datasets drawn from public archives, producing results that independently validate the framework's capabilities on real-world data at scale. Across six deployments, the system processed more than 12 000 individual objects from four independent archives (NASA Exoplanet Archive, Pantheon+, SDSS, Gaia EDR3). In each case, Astra operated without prior guidance on what relationships to seek; the system ingested the data, autonomously identified patterns, applied causal and physical reasoning, and returned interpretations that we compare with established results.

9.1 Kepler's Third Law Recovery

Astra analysed 2 839 confirmed exoplanets from the NASA Exoplanet Archive with measured orbital periods, semi-major axes,

and host-star masses. Operating in discovery mode, the system autonomously identified a power-law relationship between orbital period P and semi-major axis a, recovering the multivariate fit

$$\log P = 1.497 \log a - 0.474 \log M_\star, \tag{3}$$

with a coefficient of determination $R^2 = 0.9982$. The semi-major axis exponent of 1.497 is within 0.2 per cent of the Keplerian value of $3/2$, and the stellar-mass exponent of -0.474 is within 5.2 per cent of the theoretical $-1/2$. This multivariate form captures the stellar-mass dependence predicted by Newton's generalization of Kepler's law ($P^2 = 4\pi^2 a^3/(GM_\star)$), recovering a tighter fit than the univariate $P^2 \propto a^3$ relation. The dimensional analysis module independently validated the result by confirming consistency with Newtonian gravity. This demonstrates Astra's ability to rediscover fundamental physical laws directly from observational data, including the identification of relevant controlling variables.

9.2 Accelerating Expansion (Dark Energy Signature)

From 1 701 Type Ia supernovae in the Pantheon+ compilation (Brout et al. 2022), Astra fitted a distance modulus–redshift relation

$$\mu = 5.33 \log_{10}(z) + 24.42, \tag{4}$$

with $\chi^2/\mathrm{dof} = 0.64$, indicating an excellent fit. At redshifts $z > 0.5$, the measured luminosity distances systematically exceed the predictions of an empty (Milne) universe, recovering the observational signature of cosmic acceleration first reported by Riess et al. (1998) and Perlmutter et al. (1999). Astra flagged this deviation as statistically significant and generated the physical interpretation that an additional energy component (consistent with dark energy or a cosmological constant) is required. This test validates the framework's ability to identify subtle systematic trends in noisy data and connect them to fundamental physics.

9.3 Galaxy Colour Bimodality

Analysis of 2 000 SDSS galaxies revealed a bimodal distribution in rest-frame $u-g$ colour, with GMM-fitted peaks at $u-g = 1.44$ (blue cloud) and $u-g = 1.92$ (red sequence), consistent with the colour separation reported by Strateva et al. (2001). Astra identified the bimodality autonomously, correctly interpreting it as reflecting two distinct stellar-population histories: ongoing star formation in blue-cloud galaxies and quenched, passively evolving stellar populations in the red sequence. The blue fraction declines from ~50 per cent at low stellar masses ($\log M_\star/M_\odot \lesssim 10$) to ~10 per cent at high masses ($\log M_\star/M_\odot \gtrsim 11$), reflecting the well-known mass-dependent quenching of star formation (Peng et al. 2010). This result validates Astra's unsupervised pattern recognition on complex, multi-dimensional datasets.

9.4 Confounder Detection

As part of the galaxy colour analysis, Astra's causal reasoning module performed automated confounder detection on the SDSS galaxy sample. The system identified u-band apparent magnitude as the strongest confounder influencing the colour–mass relationship, with a measured bias of 0.2351. This is physically expected: u-band flux is sensitive to both recent star formation (which drives blue colours) and distance (which affects apparent magnitude at fixed luminosity), making it a confounding variable in any analysis that does not explicitly control for survey depth. The automated detection of this confounder demonstrates Astra's ability to identify variables that could bias scientific conclusions if left uncontrolled.

9.5 HR Diagram and the Main Sequence

Using 4 984 stars from Gaia Early Data Release 3, Astra recovered the main-sequence relationship $M_G = 3.66 \times (G_{\mathrm{BP}} - G_{\mathrm{RP}})$ with a Pearson correlation $r = 0.90$ ($p < 10^{-100}$). The physics engine provided an interpretation via the Stefan–Boltzmann law: more luminous stars have higher effective temperatures, producing bluer colours and brighter absolute magnitudes. By recovering this textbook relationship from raw photometric data, Astra demonstrates that its statistical and physical reasoning modules operate correctly on large stellar samples.

9.6 Causal Direction Discovery

Applying the PC and FCI (Zhang 2008) causal discovery algorithms to a combined stellar–galaxy dataset, Astra correctly identified redshift as a cause of observed colour (cosmological reddening), rather than the reverse. This is a non-trivial result: in purely correlational analyses, redshift and colour are symmetric, but causal algorithms break this symmetry by exploiting conditional independence structure. Astra's correct recovery of the causal direction—cosmological effects driving observed photometric properties, not the reverse—validates the causal reasoning module on real multi-source data.

9.7 Galaxy Survey Discovery Mode

As an additional deployment, Astra was applied in discovery mode to a realistic galaxy survey of 3 000 galaxies with properties drawn from SDSS scaling relations (Elbaz et al. 2007; Mannucci et al. 2010). Operating without prior hypotheses, Astra recovered: (i) the star-forming main sequence ($\log \mathrm{SFR} = 0.59 \log M_* - 5.45$, $r^2 = 0.72$); (ii) the mass–metallicity relation; and (iii) identified 150 outlier galaxies (5 per cent) warranting follow-up investigation through multi-dimensional anomaly detection. This demonstrates Astra's capacity for automated pattern discovery and triage in survey-scale datasets. Full scripts and results are available in the online supplementary material (Example 6).

These six live-data results, obtained without manual tuning or domain-specific guidance, demonstrate that Astra generalises beyond its controlled test cases. The recovered relationships span classical mechanics (Kepler's law), cosmology (dark energy), galaxy evolution (colour bimodality and survey-scale discovery), stellar astrophysics (HR diagram), and causal inference (redshift–colour direction), collectively exercising the full breadth of the framework's analytical capabilities. (Confounder detection in Section 9.4 is presented as a sub-analysis of the galaxy colour deployment rather than as an independent deployment.)

10 CROSS-DOMAIN SCIENTIFIC VALIDATION

A critical test for any general-purpose analytical framework is whether its capabilities transfer beyond its original domain. To evaluate this, we deployed Astra on publicly available datasets from three non-astrophysical domains—economics, climate science, and epidemiology—and performed a cross-domain synthesis linking

variables across these fields. In each case, Astra operated without domain-specific guidance: the system ingested the data, autonomously identified patterns, applied its causal and statistical reasoning modules, and generated hypotheses that we compare with established results.

All data used in this section are drawn from public archives: the World Bank Open Data repository (World Bank 2024), NASA Goddard Institute for Space Studies (GISS) global temperature records (Hansen et al. 2010), and the NOAA Mauna Loa CO_2 time series (NOAA Global Monitoring Laboratory 2024).

10.1 Economics: World Bank Macroeconomic Indicators

Astra analysed macroeconomic indicators for 217 countries from the World Bank Open Data repository (World Bank 2024), spanning 1960–2023 and covering GDP, unemployment, trade balance, inflation, government debt, income inequality (Gini coefficient), and related variables. Operating in discovery mode, the system generated 10 hypotheses and tested each against the data.

Validated hypotheses (9/10):

(i) **Okun's Law:** Astra recovers a negative relationship between GDP growth and unemployment change, consistent with the empirical regularity reported by Okun (1962). The estimated coefficient (−1.8 per cent unemployment per 1 per cent GDP shortfall) falls within the range reported in the literature.

(ii) **Trade–GDP correlation:** Trade openness (exports + imports as a fraction of GDP) is positively correlated with GDP per capita ($r = 0.41$, $p < 10^{-5}$), consistent with standard growth models.

(iii) **Gini inequality persistence:** Income inequality exhibits strong temporal persistence ($r = 0.92$ at 5-year lag), consistent with structural theories of inequality.

(iv) **PPP convergence:** Purchasing power parity shows convergence over decadal time-scales, consistent with the Balassa–Samuelson effect.

(v) **GDP mean-reversion:** GDP growth rates exhibit mean-reversion on 5–10 year time-scales, consistent with business cycle theory.

(vi) **Reinhart–Rogoff debt threshold:** Astra identifies a weakly negative relationship between government debt-to-GDP ratio and growth above the ∼90 per cent threshold reported by Reinhart & Rogoff (2010), though the effect is not statistically robust ($p = 0.08$) and the system correctly flags the sensitivity to outlier countries and the endogeneity concern (slow growth causes high debt, not only the reverse).

(vii) **Export diversification:** Economies with more diversified export baskets show lower GDP volatility ($r = -0.38$, $p < 0.001$).

(viii) **Inflation persistence:** Inflation rates exhibit strong autocorrelation ($r = 0.78$ at 1-year lag), consistent with adaptive expectations models.

(ix) **Finance–growth nexus:** Financial sector depth (domestic credit to private sector as a fraction of GDP) is positively associated with GDP per capita ($r = 0.52$, $p < 10^{-8}$), though the causal direction remains ambiguous.

Failed hypothesis (1/10):

(x) **Phillips Curve:** Astra tested for a negative relationship between unemployment and inflation, as originally reported by Phillips (1958). The system found no statistically significant relationship in the post-1990 cross-country data ($r = -0.03$, $p = 0.62$). This is consistent with the well-documented breakdown of the simple Phillips Curve in modern macroeconomic data and demonstrates that Astra correctly identifies non-relationships rather than forcing spurious patterns.

The Phillips Curve failure is a particularly important result: it demonstrates that Astra does not simply confirm every hypothesis it generates. The system's ability to identify relationships that do *not* hold in the data is as scientifically valuable as its ability to identify those that do.

10.2 Climate Science: NASA GISS and NOAA Data

Astra analysed global mean surface temperature anomalies from NASA GISS (Hansen et al. 2010) and atmospheric CO_2 concentrations from the NOAA Mauna Loa Observatory (NOAA Global Monitoring Laboratory 2024), covering the period 1880–2023 (temperature) and 1958–2023 (CO_2).

Validated hypotheses (4/5):

(i) **CO_2–temperature correlation:** Astra identifies a strong positive correlation between atmospheric CO_2 concentration and global mean temperature anomaly ($R^2 = 0.936$ over the 1958–2023 overlap period). The system correctly notes that correlation does not establish causation but flags the temporal precedence of CO_2 increases as suggestive of a causal relationship.

(ii) **Warming acceleration:** The rate of warming post-1990 ($0.20\,^\circ\mathrm{C\,decade^{-1}}$) is 67 per cent faster than the pre-1990 rate ($0.12\,^\circ\mathrm{C\,decade^{-1}}$), consistent with accelerating radiative forcing from greenhouse gas accumulation.

(iii) **CO_2 growth acceleration:** The annual rate of CO_2 increase has itself accelerated: the mean annual increment was $\sim 1.0\,\mathrm{ppm\,yr^{-1}}$ in the 1960s, rising to $\sim 2.4\,\mathrm{ppm\,yr^{-1}}$ in the 2010s, consistent with increasing anthropogenic emissions.

(iv) **Decadal warming trend:** Each successive decade since the 1970s has been warmer than the preceding one, a pattern that Astra identifies as statistically significant at the $p < 0.001$ level using a non-parametric trend test.

Inconclusive hypothesis (1/5): The system tested for an El Niño–Southern Oscillation (ENSO) modulation signal in the temperature residuals after removing the long-term trend. While periodic structure was detected, Astra could not establish a statistically robust ENSO signal with the available temporal resolution, and correctly flagged the result as inconclusive.

10.3 Epidemiology: World Bank Health Indicators

Astra analysed health and development indicators for 217 countries from the World Bank (World Bank 2024), including life expectancy, infant mortality, GDP per capita, health expenditure, vaccination coverage, and maternal mortality.

Validated hypotheses (5/5):

(i) **Infant mortality vs. GDP:** Astra identifies a strong negative relationship between infant mortality rate and GDP per capita ($R^2 = 0.587$), with a logarithmic functional form providing the best fit—consistent with diminishing marginal returns of wealth on health outcomes.

(ii) **Preston Curve:** Life expectancy increases with GDP per capita following a concave (logarithmic) relationship ($R^2 = 0.686$), recovering the classic Preston Curve (Preston 1975). The system correctly notes the flattening at high GDP, indicating saturation of the wealth–health relationship.

(iii) **Life expectancy vs. health spending:** Health expenditure per capita is positively associated with life expectancy ($r = 0.62$, $p < 10^{-12}$), though Astra's causal module flags the likely bidirectional relationship (wealthier countries both spend more on health and have longer life expectancies, with GDP as a common cause).

(iv) **DPT vaccination coverage:** DPT3 vaccination coverage is negatively correlated with infant mortality ($r = -0.65$, $p < 10^{-15}$), consistent with the established effectiveness of childhood vaccination programmes.

(v) **Maternal mortality:** Maternal mortality ratio is strongly negatively associated with GDP per capita ($r = -0.58$, $p < 10^{-10}$) and with skilled birth attendance ($r = -0.71$, $p < 10^{-18}$), consistent with the role of healthcare infrastructure in reducing maternal deaths.

10.4 Cross-Domain Synthesis

To test whether Astra can identify meaningful relationships *across* domains, we merged the economics, climate, and health datasets at the country–year level and applied the full analytical pipeline. The system generated 8 cross-domain hypotheses and applied Benjamini–Hochberg false discovery rate (FDR) correction (Benjamini & Hochberg 1995) at $q = 0.05$ to account for multiple testing.

Validated after FDR correction (5/8):

(i) **GDP–CO_2 coupling:** National GDP is positively correlated with per-capita CO_2 emissions ($r = 0.71$, $p_{\mathrm{adj}} < 10^{-10}$), reflecting the carbon intensity of economic activity.

(ii) **Life expectancy–CO_2:** Countries with higher CO_2 emissions per capita tend to have higher life expectancies ($r = 0.58$, $p_{\mathrm{adj}} < 10^{-6}$). Astra correctly identifies this as a confounded relationship (GDP drives both) rather than a causal link from emissions to health.

(iii) **Wealth–Health Nexus:** GDP per capita, health expenditure, and life expectancy form a strongly coupled triad ($R^2 = 0.700$ for the three-variable model), which Astra identifies as the strongest cross-domain relationship. The causal module suggests GDP → health expenditure → life expectancy as the primary causal pathway, with direct GDP → life expectancy effects mediated by nutrition, sanitation, and education.

(iv) **Urbanization–CO_2:** Urbanization rate is positively correlated with per-capita CO_2 emissions ($r = 0.49$, $p_{\mathrm{adj}} < 10^{-4}$), consistent with higher energy consumption in urban settings.

(v) **Renewables–GDP paradox:** Renewable energy share is *negatively* correlated with GDP per capita ($r = -0.42$, $p_{\mathrm{adj}} < 0.001$). Astra correctly interprets this counter-intuitive finding: poorer countries rely more heavily on traditional biomass (counted as renewable), while wealthier countries have historically derived more energy from fossil fuels. The system flags this as an example where naïve interpretation of a correlation would be misleading.

Lost after FDR correction (3/8): Three hypotheses (education–emissions coupling, democracy–health association, and trade openness–environmental quality) did not survive FDR correction at $q = 0.05$. These marginal relationships may be real but require larger samples or more careful confound control to establish.

The application of FDR correction is methodologically important: without it, the cross-domain analysis would have reported 8/8 validated hypotheses, a misleadingly optimistic result. Astra's automatic application of FDR correction demonstrates responsible statistical practice.

10.5 Negative Results and Epistemic Honesty

The cross-domain validation provides several examples of Astra correctly identifying non-relationships or qualifying ambiguous findings:

Phillips Curve failure: As discussed in Section 10.1, Astra found no statistically significant unemployment–inflation trade-off in post-1990 data ($r = -0.03$, $p = 0.62$), consistent with the well-documented instability of the Phillips Curve.

FDR attrition: Of 8 cross-domain hypotheses, 3 were lost after FDR correction. Astra reports these as "not validated" rather than attempting to salvage them through post-hoc analysis.

Confound identification: In 3 of the 5 validated cross-domain hypotheses, Astra's causal module identified confounding variables (primarily GDP as a common cause) and flagged the correlations as potentially non-causal.

Reinhart–Rogoff qualification: The debt–growth relationship was validated only with substantial caveats ($p = 0.08$, sensitivity to outliers, endogeneity concerns), demonstrating nuanced rather than binary hypothesis assessment.

These negative and qualified results are as important as the positive validations. A system that confirms every hypothesis it generates would be useless; Astra's ability to distinguish signal from noise, and to qualify ambiguous findings, is essential for its role as a scientific tool.

10.6 Summary of Cross-Domain Results

Table 2 provides a comprehensive listing of all hypotheses tested across the five scientific domains.

*Reinhart–Rogoff threshold validated with substantial caveats: $p = 0.08$, sensitive to outliers, endogeneity concerns flagged.

In total, Astra tested approximately 43 hypotheses across five scientific domains, validating 38 (including 1 with caveats), failing 1 (Phillips Curve), finding 1 inconclusive (ENSO), and losing 3 after FDR correction. This 88 per cent validation rate—achieved without domain-specific tuning and across disciplines as diverse as astrophysics, macroeconomics, and epidemiology—provides strong evidence that Astra's analytical capabilities generalize beyond its original astrophysical domain.

11 DISCUSSION

11.1 What ASTRA Demonstrates

The six controlled astrophysical test cases, six live deployments, cross-domain validation, and stigmergic swarm intelligence collectively analyse data drawn from more than a dozen independent sources spanning five scientific domains, with 397 total discoveries and 38 validated hypotheses. The astrophysical analyses process more than 27 430 data points from nine sources, while the cross-domain validation draws on national-level panel data for 217 countries over six decades (World Bank), global temperature records spanning 143 years (NASA GISS), and atmospheric CO_2 measurements covering 65 years (NOAA Mauna Loa). Rather than repeating individual results, we reflect here on what these collective demonstrations reveal about Astra's strengths and limitations.

The integration advantage is most concretely demonstrated in Test Case 5 (Bayesian Model Selection), where physical motivation (the virial theorem predicting a power law) and statistical evidence (Bayes factors strongly favouring power-law and logarithmic models over

Table 2. Summary of all hypotheses tested across five scientific domains. "Validated" indicates the hypothesis is supported by the data at the stated significance level; "Failed" indicates the hypothesis is not supported; "FDR lost" indicates the hypothesis did not survive false discovery rate correction.

Domain	Hypothesis	Key Statistic	Valid?	Data Source
Astrophysics — Controlled Test Cases				
Astrophysics	Filament universal width	0.098 ± 0.019 pc	✓	Herschel
Astrophysics	Virial scaling relation	$r = 0.988$, $p < 10^{-18}$	✓	Herschel
Astrophysics	Multi-λ cross-match	60/370 secure matches	✓	CDFS
Astrophysics	Mass–metallicity relation	$r = 0.68$, $p < 10^{-27}$	✓	SDSS
Astrophysics	Environmental quenching	ΔSFR ≈ -0.3 dex	✓	SDSS
Astrophysics	Causal structure (Gaia)	Correct DAG recovery	✓	Gaia DR2
Astrophysics	Bayesian model selection	BF = 1.2 (PL vs. log)	✓	Herschel
Astrophysics	SF causal drivers (synthetic)	3/3 drivers found	✓	Synthetic
Astrophysics	Column density as proxy	Proxy confirmed	✓	Synthetic
Astrophysics — Live Deployments				
Astrophysics	Kepler's third law	$R^2 = 0.9982$	✓	NASA Exo. Archive
Astrophysics	Dark energy signature	$\chi^2/\mathrm{dof} = 0.64$	✓	Pantheon+
Astrophysics	Galaxy colour bimodality	Peaks: 1.44, 1.92	✓	SDSS
Astrophysics	u-band confounder	Bias = 0.2351	✓	SDSS
Astrophysics	HR main sequence	$r = 0.90$, $p < 10^{-100}$	✓	Gaia EDR3
Astrophysics	Causal direction ($z \to$ colour)	Correct orientation	✓	Gaia+SDSS
Economics				
Economics	Okun's Law	$-1.8\%/1\%$ GDP	✓	World Bank
Economics	Trade–GDP correlation	$r = 0.41$, $p < 10^{-5}$	✓	World Bank
Economics	Gini persistence	$r = 0.92$ (5-yr lag)	✓	World Bank
Economics	PPP convergence	Decadal convergence	✓	World Bank
Economics	GDP mean-reversion	5–10 yr cycle	✓	World Bank
Economics	Reinhart–Rogoff threshold	$p = 0.08$ (qualified)	✓*	World Bank
Economics	Export diversification	$r = -0.38$, $p < 0.001$	✓	World Bank
Economics	Inflation persistence	$r = 0.78$ (1-yr lag)	✓	World Bank
Economics	Finance–growth nexus	$r = 0.52$, $p < 10^{-8}$	✓	World Bank
Economics	Phillips Curve	$r = -0.03$, $p = 0.62$	×	World Bank
Climate Science				
Climate	CO_2–temp. correlation	$R^2 = 0.936$	✓	GISS + NOAA
Climate	Warming acceleration	67% faster post-1990	✓	GISS
Climate	CO_2 growth acceleration	$1.0 \to 2.4$ ppm yr^{-1}	✓	NOAA
Climate	Decadal warming trend	$p < 0.001$	✓	GISS
Climate	ENSO modulation	Inconclusive	–	GISS
Epidemiology				
Epidemiology	Infant mortality vs. GDP	$R^2 = 0.587$	✓	World Bank
Epidemiology	Preston Curve	$R^2 = 0.686$	✓	World Bank
Epidemiology	Life exp. vs. health spending	$r = 0.62$, $p < 10^{-12}$	✓	World Bank
Epidemiology	DPT vaccination effect	$r = -0.65$, $p < 10^{-15}$	✓	World Bank
Epidemiology	Maternal mortality	$r = -0.71$, $p < 10^{-18}$	✓	World Bank
Cross-Domain Synthesis (after FDR correction)				
Cross-domain	GDP–CO_2 coupling	$r = 0.71$, $p_{\mathrm{adj}} < 10^{-10}$	✓	WB + NOAA
Cross-domain	Life exp.–CO_2 (confounded)	$r = 0.58$, $p_{\mathrm{adj}} < 10^{-6}$	✓	WB + NOAA
Cross-domain	Wealth–Health Nexus	$R^2 = 0.700$	✓	World Bank
Cross-domain	Urbanization–CO_2	$r = 0.49$, $p_{\mathrm{adj}} < 10^{-4}$	✓	WB + NOAA
Cross-domain	Renewables–GDP paradox	$r = -0.42$, $p_{\mathrm{adj}} < 0.001$	✓	World Bank
Cross-domain	Education–emissions	Not significant after FDR	FDR lost	WB + NOAA
Cross-domain	Democracy–health	Not significant after FDR	FDR lost	World Bank
Cross-domain	Trade–environment	Not significant after FDR	FDR lost	World Bank

linear alternatives) jointly inform model interpretation within a single automated workflow. This joint consideration of statistical evidence and physical constraints—without requiring manual synthesis of separate analyses—represents the core value of the integrated approach.

Test Case 6 demonstrates a distinct capability: discovery-mode operation beyond knowledge retrieval. By operating in "knowledge isolation mode" on a synthetic dataset with embedded causal structure, Astra discovered the correct causal drivers, identified column density as a proxy rather than direct cause, and generated testable predictions. This addresses a key question: *can Astra go beyond "correct recovery of known results" to demonstrate genuine discovery capability?* The results suggest that Astra's architecture supports discovery-mode operation, though definitive scientific discovery requires collaboration with domain experts and observational validation.

Four Modes of Evidence:

(i) **Validation mode** (Tests 1–5): Astra analyses real data to

recover known astrophysical relationships, validating the integrated framework.

(ii) **Discovery mode** (Test 6): ASTRA analyses data in knowledge isolation mode to discover patterns without being told what to look for, generating hypotheses and testable predictions.

(iii) **Live deployment** (Section 9): ASTRA operates on archival data without manual guidance, independently recovering fundamental physics.

(iv) **Cross-domain generalization** (Section 10): ASTRA operates on non-astrophysical data, recovering established relationships and correctly identifying non-relationships—demonstrating that the framework's capabilities are not domain-specific artefacts.

11.2 What ASTRA Genuinely Contributes

A fair question is whether ASTRA merely confirms what is already known. The answer is nuanced.

The live validation results, while impressive for a fully automated system, recover well-established physics. Their value lies in demonstrating that ASTRA's integrated pipeline produces correct, quantitatively precise results on real data without human intervention. That an automated system can ingest raw archival data and arrive at $R^2 = 0.9982$ for Kepler's law or $\chi^2/\mathrm{dof} = 0.64$ for the Hubble diagram is a non-trivial engineering and methodological achievement, even though the underlying physics is textbook material.

The cross-domain results extend this argument: recovering Okun's Law, the Preston Curve, and CO_2–temperature coupling using the same pipeline that recovers Kepler's law demonstrates that ASTRA's capabilities are genuinely domain-general. The Phillips Curve failure and FDR attrition provide further evidence that the system produces honest, calibrated results rather than indiscriminate confirmations.

ASTRA's genuine scientific contribution lies in four areas. First, the *integration itself*: by chaining dimensional analysis, causal inference, and Bayesian model selection into a single workflow, ASTRA eliminates inconsistencies that arise when scientists manually combine outputs from separate tools—particularly in uncertainty propagation and physical validation. Second, the *causal reasoning capability*: Test Case 6 demonstrates that ASTRA can distinguish proxy variables from causal drivers, a distinction that purely correlational or ML-based approaches cannot make. Third, the *knowledge isolation mode*: by disabling access to encoded domain knowledge, ASTRA provides a controlled test of whether patterns in data are discoverable from first principles, an approach that could be applied to domains where the ground truth is genuinely unknown. Fourth, the *cross-domain generalization*: the same analytical pipeline that recovers astrophysical scaling relations also recovers macroeconomic regularities, climate trends, and epidemiological relationships, with appropriate epistemic qualifications.

11.3 Cross-Domain Generalization

The cross-domain validation (Section 10) addresses a fundamental question: is the framework genuinely general-purpose, or are its capabilities specific to the astrophysical domain for which it was developed?

Three aspects of the results support generality. First, the *same analytical modules*—dimensional analysis, causal inference, Bayesian model selection, pattern recognition—that recover astrophysical relationships also recover economic, climatic, and epidemiological relationships, without domain-specific modifications. Second, ASTRA's *epistemic qualifications* transfer across domains: the system correctly identifies GDP as a confound in the life expectancy–CO_2 correlation, flags the Reinhart–Rogoff threshold as sensitive to outliers and endogeneity, and applies FDR correction to the cross-domain synthesis. Third, the *negative results*—the Phillips Curve failure, the ENSO inconclusive result, and the three FDR-lost cross-domain hypotheses—demonstrate that ASTRA does not indiscriminately confirm hypotheses.

These results suggest that ASTRA's architecture is genuinely domain-agnostic at the analytical level. The core capabilities—pattern discovery, causal inference, hypothesis testing with multiple testing correction, confound identification—are domain-general by construction.

Caveats: The cross-domain validation uses well-established relationships as benchmarks. ASTRA has not been tested on genuinely open questions in non-astrophysical domains. The domain knowledge systems (MORK ontology, 75 specialized modules) are astrophysics-specific; extending to production-quality analysis in other domains would require developing analogous domain knowledge. The cross-domain results demonstrate that the *analytical architecture* generalizes, not that ASTRA is ready for deployment as an economics or climate science tool.

11.4 Cognitive Architecture and Multi-Agent Reasoning

The cognitive architecture (Section 2.8), multi-agent debate system (Section 2.9), theory framework (Section 2.10), and autonomous research agenda (Section 2.11) represent a significant extension of ASTRA's capabilities beyond the core analytical pipeline validated in the test cases. These components are designed to address limitations of purely pipeline-based scientific analysis: the knowledge graph maintains persistent context across analyses; neuro-symbolic integration provides formal verification of statistically discovered patterns; multi-agent debate reduces confirmation bias through structured adversarial evaluation; and the theory framework supports the progression from individual validated hypotheses to broader theoretical understanding.

We emphasize that these components have not been independently validated through controlled experiments in this paper. The test case results reported in Sections 3–8 and the live deployments in Section 9 exercise the core analytical modules (dimensional analysis, causal inference, Bayesian model selection) rather than the cognitive architecture or multi-agent debate system in isolation. Rigorous ablation studies—comparing analytical performance with and without the cognitive architecture, knowledge graph, and multi-agent debate—are required to quantify their individual contributions and are planned for future work.

The autonomous research agenda, with its information-theoretic curiosity metrics and human approval gates, provides a principled framework for directing computational effort toward scientifically meaningful questions. However, the effectiveness of curiosity-driven exploration relative to human-directed analysis remains an open empirical question.

11.5 Positioning Relative to Other Approaches

ASTRA occupies a distinct position in the computational astronomy ecosystem by integrating multiple analysis types within a unified framework.

Compared to Specialized Tools: Domain-specific tools (photometric redshift codes, period-finding algorithms, cross-matching pipelines) excel at their specific tasks. ASTRA does not aim to re-

place them but to provide integrated workflows that combine multiple analysis types with consistent uncertainty handling and physical validation.

Compared to Manual Pipelines: Astronomers routinely combine multiple tools manually, creating opportunities for inconsistent uncertainty treatment and reproducibility challenges. Astra addresses these by providing automated workflows with integrated validation.

Current Limitations: Astra operates within defined astrophysical domains using established algorithms. The system assists scientific reasoning rather than replacing domain expertise. The test cases presented use small samples and should be understood as capability demonstrations rather than production-scale analyses.

11.6 ASTRA's Role in Scientific Discovery

A crucial question is: *what role can Astra play in genuine scientific discovery?* We would not expect to give Astra one of the big contemporary problems—"resolve the nature of the Hubble Tension," "tell us what dark matter is," "prove that we live in a holographic universe"—and receive definitive answers in a computational Eureka moment. These problems require deep physical insight, creative hypothesis formation, and careful experimental design—capabilities that remain fundamentally human.

However, Astra's discovery architecture, used alongside an experienced astronomer, provides capabilities that go beyond those of straightforward AI or machine learning. Test Case 6 demonstrates that Astra can discover patterns in data without being told what to look for (knowledge isolation mode), apply causal inference methods that distinguish correlation from causation, generate competing hypotheses with quantitative rankings, and produce testable predictions to guide future observations.

Astra's role is to work alongside the experienced astronomer to analyse data and facilitate genuine discovery. The human scientist brings deep physical understanding, creative insight, and judgment about which questions are worth asking. Astra brings automated pattern discovery, causal reasoning, and systematic hypothesis evaluation. Together, they can pursue scientific questions more effectively than either could alone.

This collaborative model—Astra as discovery assistant working alongside domain experts—is the appropriate framing for AI-assisted scientific discovery. The live dashboard (Section 2.6) operationalizes this model by providing real-time intervention controls and alignment monitoring, ensuring that the human operator retains meaningful oversight throughout autonomous discovery cycles. The stigmergic swarm intelligence layer (Section 2.7) further enhances this collaboration by autonomously balancing exploration of novel scientific domains against exploitation of productive directions, guided by biologically-calibrated parameters (Fig. 3) and validated through A/B testing against unguided baselines. Claims that AI systems will autonomously resolve major open questions overlook the essential role of human physical insight and careful experimental design.

11.7 Limitations and Future Work

Current limitations include:

- **Sample sizes:** The controlled test cases use 24–1000 objects. The live deployments reach larger scales (up to 4 984 stars), but scaling to survey-scale datasets (10^6–10^9 objects) requires further validation.
- **Computational cost:** Causal discovery scales as $O(p^2)$ for p variables, limiting applicability to high-dimensional problems.
- **Prior dependence:** Bayesian evidence computation requires careful prior specification, and results can be prior-dependent.
- **Domain knowledge:** Astra's pattern recognition retrieves encoded domain knowledge. Genuine novel discovery requires identifying patterns not already in the knowledge base.
- **Selection effects:** Multi-wavelength cross-matching assumptions (Gaussian errors, uniform priors) may not hold in all regimes.
- **Live validation scope:** The live deployment results recover well-established physics and validate the pipeline's correctness, not its ability to find genuinely new science.
- **Cross-domain depth:** The cross-domain validation uses well-established benchmarks. Production-quality analysis in other domains would require domain-specific knowledge modules analogous to Astra's astrophysical MORK ontology.
- **Cognitive architecture:** The cognitive architecture, multi-agent debate, theory framework, and autonomous research agenda (Sections 2.8–2.11) are operational but have not been independently validated through ablation studies. Their individual contributions to analytical quality remain to be quantified.

Astra is currently being applied to ongoing research on interstellar medium filaments, where its dimensional analysis and causal inference capabilities are being used to investigate filament stability and fragmentation properties.

Future work should focus on: scaling to larger datasets; validation on more scientifically challenging problems where the answer is not known in advance; integration with time-domain surveys for real-time analysis; extension to gravitational wave and neutrino multi-messenger astronomy; applying knowledge-isolation discovery mode to real datasets where causal structure is not known a priori; developing domain knowledge modules for non-astrophysical fields; and rigorous evaluation of the stigmergic swarm intelligence layer through controlled experiments comparing pheromone-guided and unguided discovery on standardized benchmarks with sufficient statistical power.

12 CONCLUSION

We have presented Astra (Autonomous System for Scientific Discovery in Astrophysics), an integrated framework for physics-aware scientific analysis that unifies dimensional analysis, causal reasoning, Bayesian model selection, multi-wavelength data fusion, and stigmergic swarm intelligence within a single reproducible pipeline. Through six controlled astrophysical test cases, six live deployments on archival data, and cross-domain validation across economics, climate science, and epidemiology, we have demonstrated Astra's technical capabilities across 38 validated hypotheses in five scientific domains, with a stigmergic swarm intelligence layer coordinating autonomous hypothesis exploration through biologically-inspired agents (Fig. 3).

The controlled test cases validate Astra's core analytical modules on real observational data. The scaling relations analysis (Test 1) recovers the ~0.1 pc filament width and virial scaling ($r = 0.988$) from 24 Herschel filaments; the multi-wavelength fusion (Test 2) achieves Bayesian cross-matching in the CDFS with a <5 per cent false match rate; and the pattern recognition (Test 3) identifies established galaxy relationships including the mass–metallicity relation and environmental quenching from 600 SDSS galaxies. The causal inference demonstrator (Test 4) correctly recovers textbook causal relationships from 1000 Gaia stars, and the Bayesian model selection (Test 5) demonstrates the automatic Occam's razor, strongly

preferring power-law and logarithmic models over linear and broken power-law alternatives.

The discovery-mode test (Test 6) provides the strongest evidence of ASTRA's analytical capabilities. Operating in knowledge isolation mode on synthetic molecular cloud data, ASTRA correctly identified all three causal drivers of star formation (Jeans mass, magnetic field, virial parameter), distinguished column density as a proxy rather than direct cause despite its highest univariate feature importance, and generated four testable predictions validated through intervention analysis.

Six live deployments on archival data independently recover established physics without domain-specific guidance: Kepler's third law from 2 839 exoplanets ($R^2 = 0.9982$), the dark energy signature from 1 701 Type Ia supernovae ($\chi^2/\mathrm{dof} = 0.64$), galaxy colour bimodality ($u-g$ peaks at 1.44 and 1.92), the HR diagram main sequence ($r = 0.90$ from 4 984 Gaia stars), correct causal direction for redshift–colour, and automated confounder detection. The cross-domain validation extends these results beyond astrophysics: ASTRA recovers 9/10 economic hypotheses (Okun's Law, trade–GDP, and seven others; Phillips Curve correctly identified as non-significant), 4/5 climate hypotheses (CO_2–temperature $R^2 = 0.936$), 5/5 epidemiological hypotheses (Preston Curve $R^2 = 0.686$), and 5/8 cross-domain syntheses surviving FDR correction.

These results establish four modes of evidence for ASTRA's capabilities: validation of known results on real data, discovery-mode operation under controlled conditions, live deployment at scale on archival data, and cross-domain generalization beyond astrophysics. While individual components use established algorithms, their integration within a single framework—coordinated by a cognitive architecture with knowledge graph persistence, neuro-symbolic verification, and multi-agent hypothesis evaluation—enables analyses that would otherwise require manual combination of multiple separate tools and pipelines. ASTRA is a tool to assist the astronomer, not a replacement for domain expertise; definitive scientific discovery requires collaboration with domain experts and observational validation.

The complete source code, documentation, and reproducible analysis notebooks are available at `https://github.com/Tilanthi/ASTRA` (White 2026).

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The stigmergic swarm intelligence layer (Section 2.7) builds upon foundational work on stigmergic intelligence and autonomous navigation by Dey (2025) at OpenHub, Thailand. OpenHub served as a research partner for this work, contributing to the development and testing of ASTRA's autonomous research capabilities through the Taurus multi-agent orchestration platform (`https://taurus.cloud`). The Taurus platform provided the computational infrastructure for ASTRA's autonomous research cycles, enabling persistent multi-agent coordination, scheduled discovery runs, and real-time hypothesis validation.

AUTHOR CONTRIBUTIONS

GJW conceived the ASTRA project, designed the framework architecture including the V8.0 cognitive architecture and V9.0 multi-agent system, led the scientific validation, and wrote the manuscript. The ASTRA codebase was developed with contributions from OpenHub (Thailand), who also conducted independent review and testing of the system—including the cognitive architecture, multi-agent debate, theory framework, and autonomous research agenda—through the Taurus research platform. Large language model assistance (Anthropic Claude) was used in manuscript preparation, including initial drafting, editing, and multiple rounds of simulated peer review. The scientific content, analysis, and conclusions were designed, executed, and verified by the author. All errors and omissions are the responsibility of the author. The author declares no conflict of interest.

DATA AVAILABILITY

The complete ASTRA source code and analysis notebooks are available at `https://github.com/Tilanthi/ASTRA`. Six fully reproducible worked examples with data generation scripts, analysis code, and figure generation are provided in the online repository (`Example1-6/`). All data used in this paper are publicly available from the original survey archives as cited in the text. World Bank data are available at `https://data.worldbank.org`; NASA GISS temperature data at `https://data.giss.nasa.gov/gistemp/`; NOAA Mauna Loa CO_2 data at `https://gml.noaa.gov/ccgg/trends/`.

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